

ECG examples - RAW (mV), as it would come off a Holter

51 example beats, grouped by MIT-BIH annotation type

Threshold (model decision): 0.0434

Distribution:

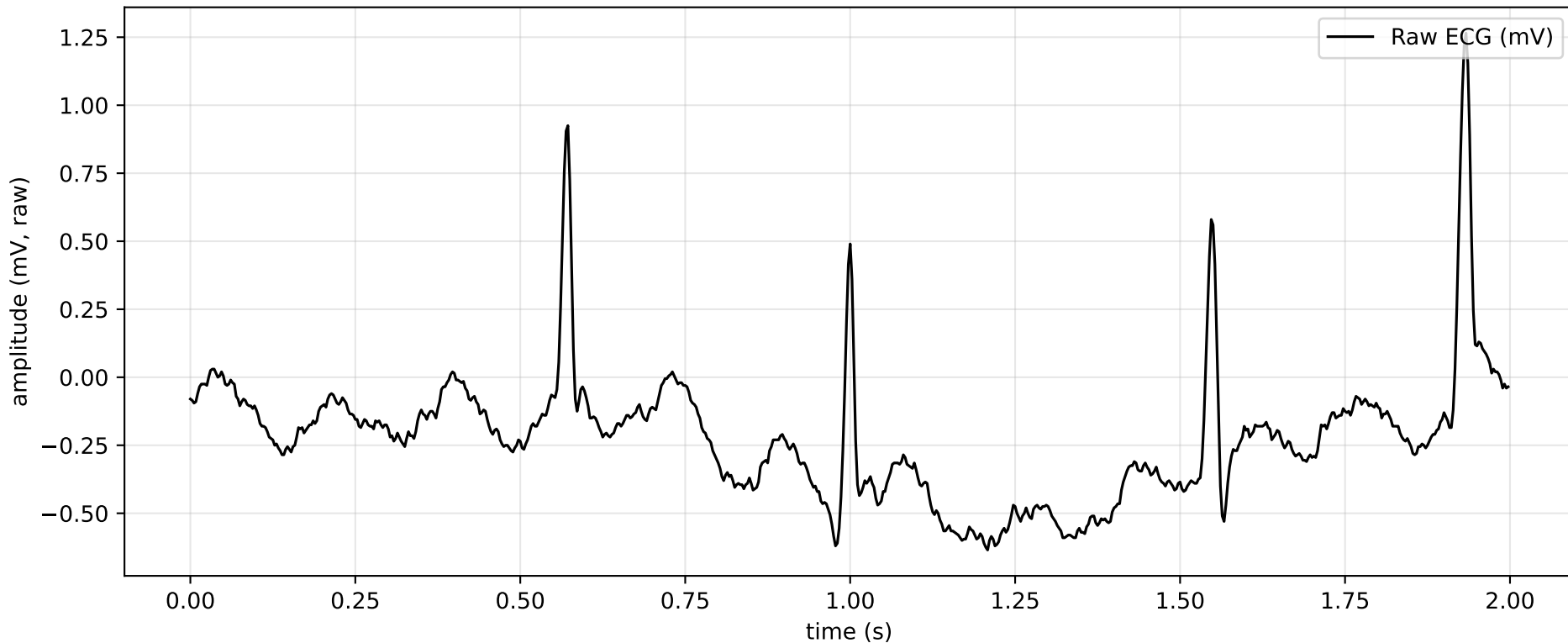
N Normal sinus beat	10
V Premature ventricular contraction (PVC)	8
F Fusion of ventricular and normal	8
L Left bundle branch block	8
A Atrial premature beat (APC)	6
R Right bundle branch block	5
/ Paced beat	4
f Fusion of paced and normal	2

Labels on each page:

Symbol	cardiologist annotation
Type	plain-language meaning
Source	MIT-BIH record + sample index (reproducible)
Residual	model's reconstruction error
Threshold	the F1-optimal value 0.0434
Decision	model's NORMAL vs ANOMALY call
Outcome	TP (caught), FN (missed), TN (correctly cleared), FP (false alarm)

Section: N - Normal sinus beat

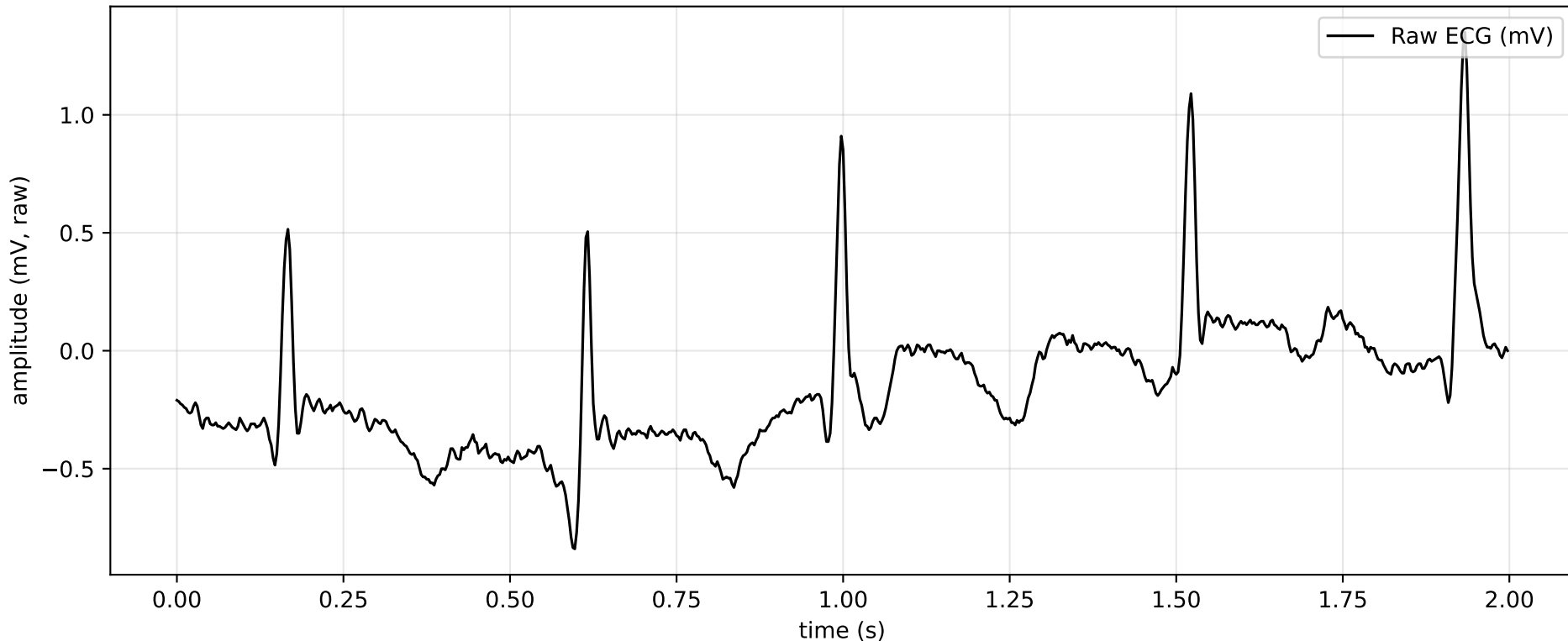
FP: Normal sinus beat (N)



Symbol	N
Type	Normal sinus beat
Source	record 222, sample 615454
Residual	1.0414
Threshold	0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	FP x mismatch

ANOMALY

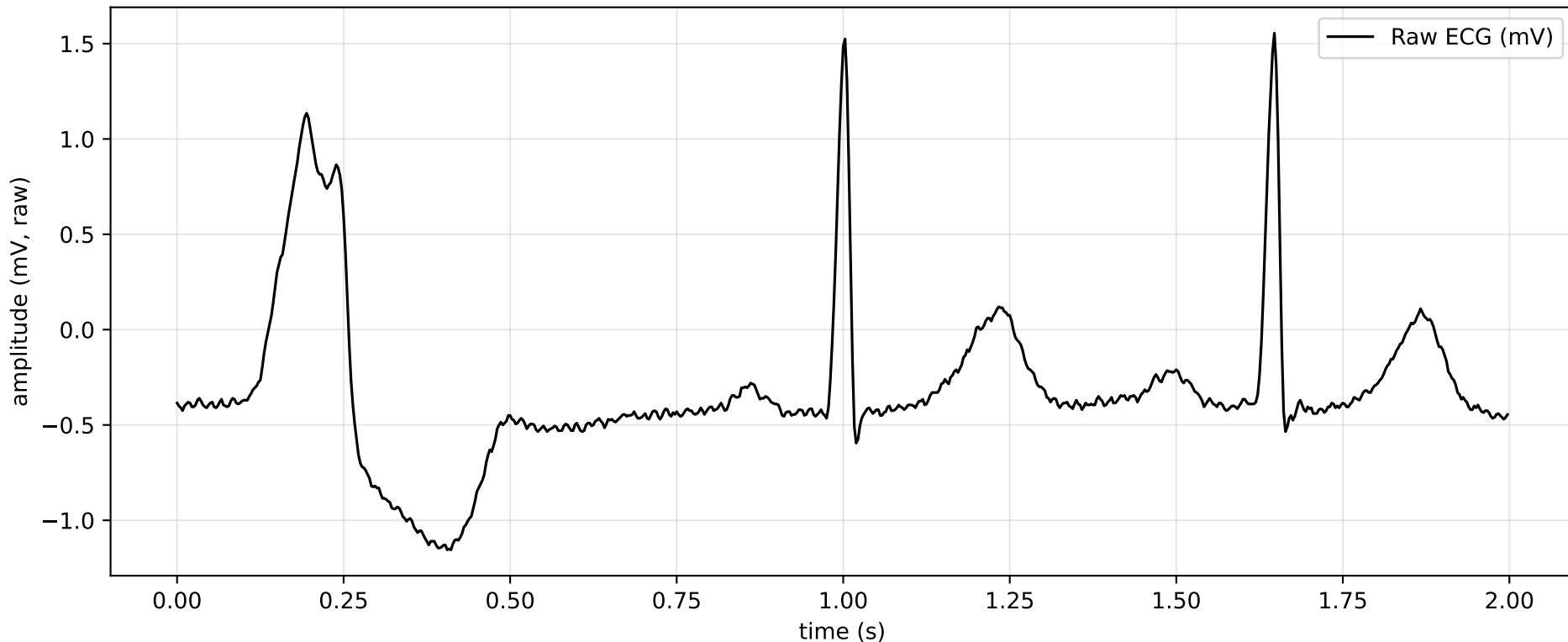
FP: Normal sinus beat (N)



Symbol	N
Type	Normal sinus beat
Source	record 222, sample 333416
Residual	1.0266
Threshold	0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	FP x mismatch

ANOMALY

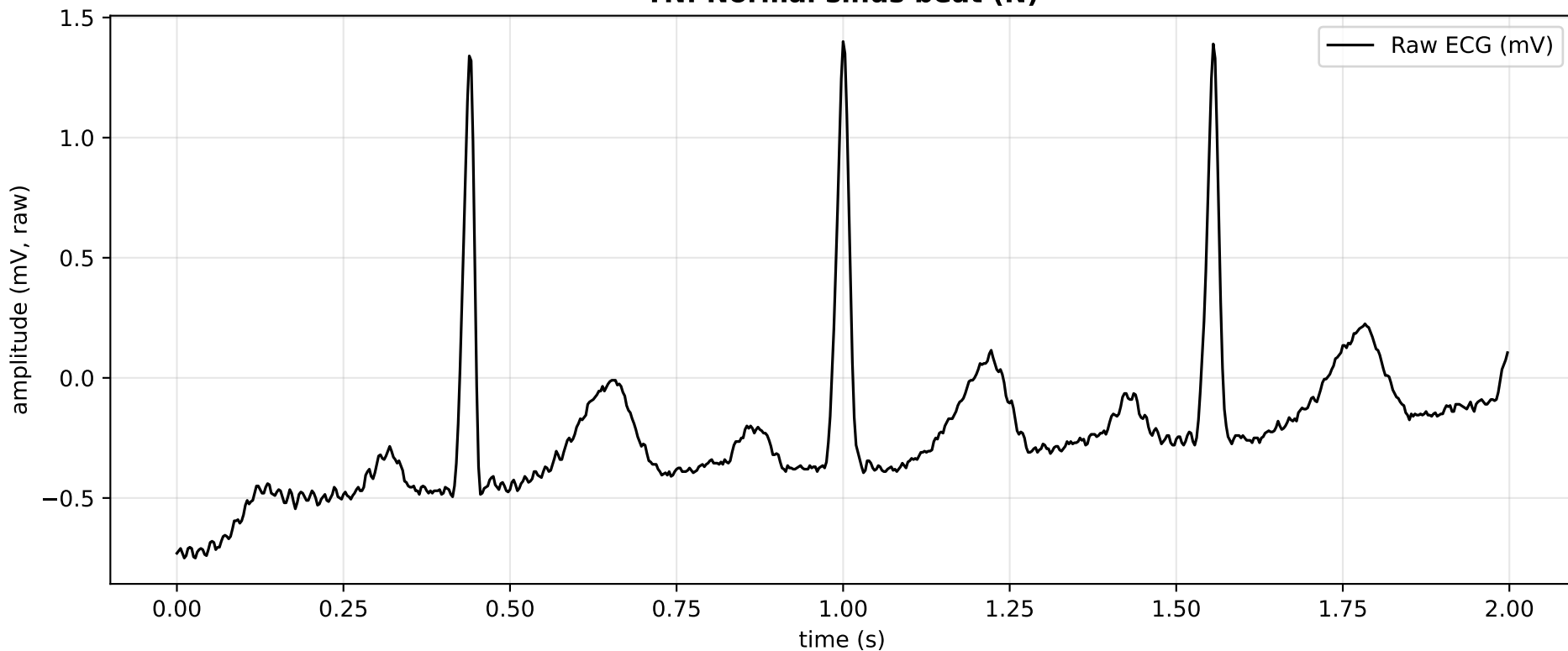
TN: Normal sinus beat (N)



Symbol	N
Type	Normal sinus beat
Source	record 208, sample 563412
Residual	0.0019
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN ✓ correct

NORMAL

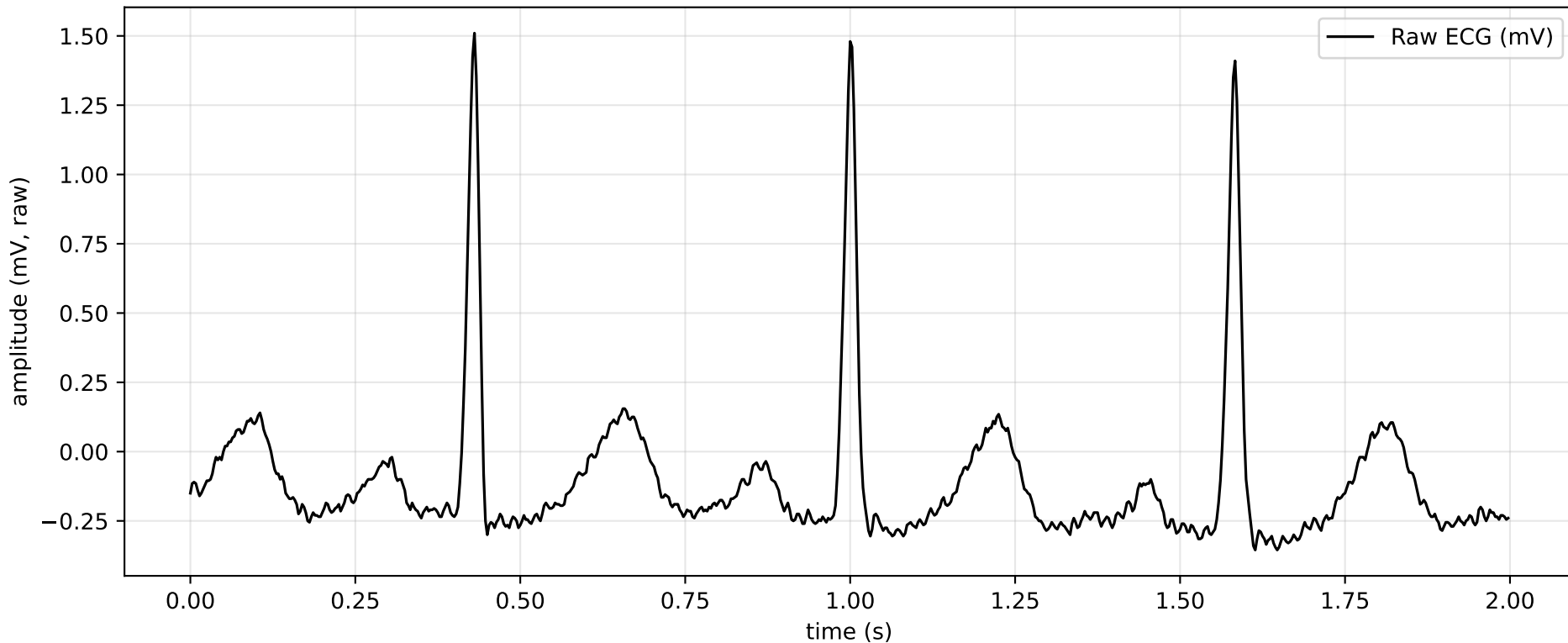
TN: Normal sinus beat (N)



Symbol	N
Type	Normal sinus beat
Source	record 208, sample 510226
Residual	0.0019
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN ✓ correct

NORMAL

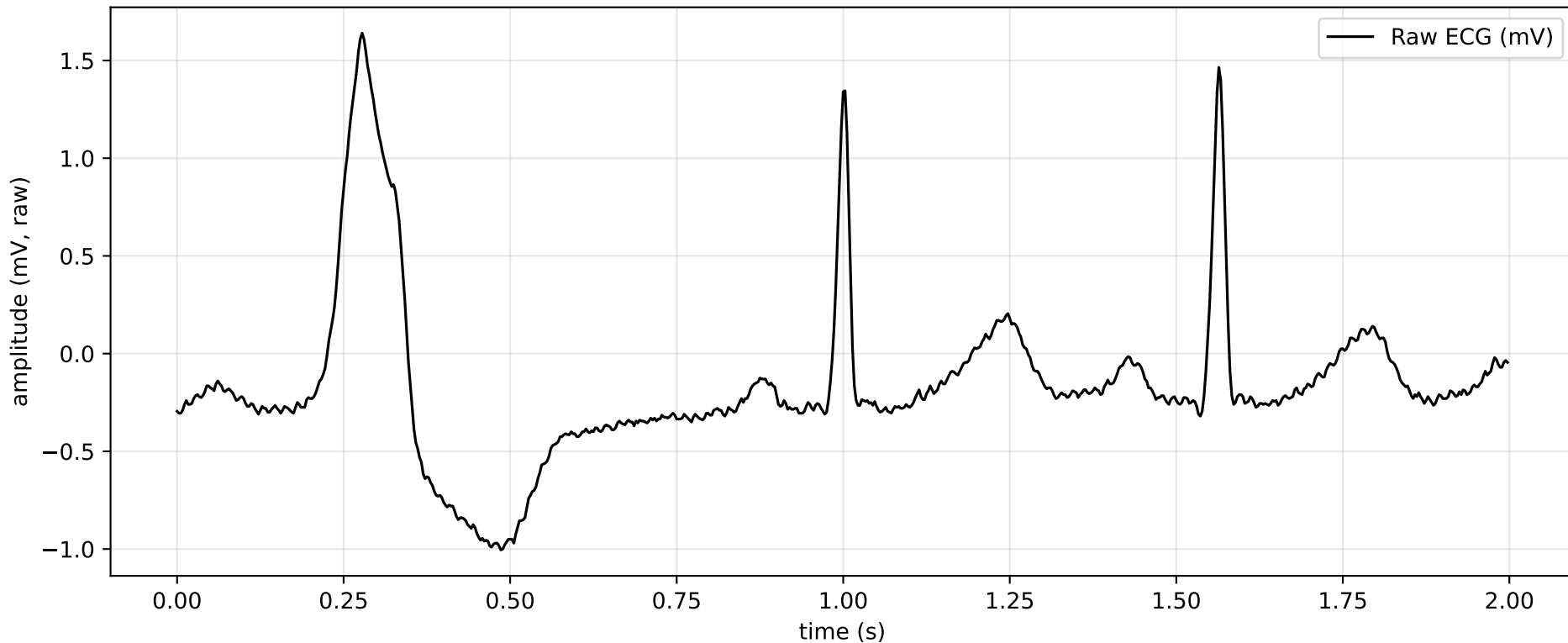
TN: Normal sinus beat (N)



Symbol	N
Type	Normal sinus beat
Source	record 208, sample 30190
Residual	0.0020
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN ✓ correct

NORMAL

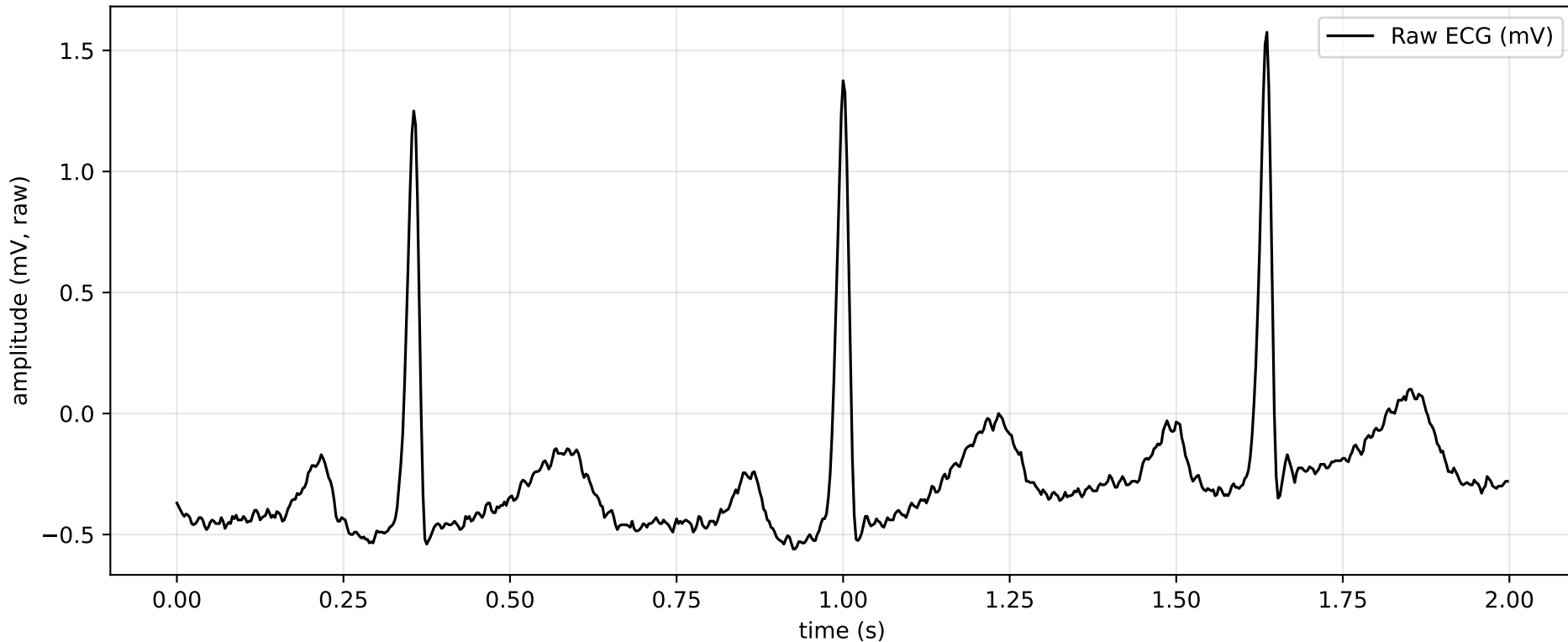
TN: Normal sinus beat (N)



Symbol	N
Type	Normal sinus beat
Source	record 208, sample 120409
Residual	0.0020
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN ✓ correct

NORMAL

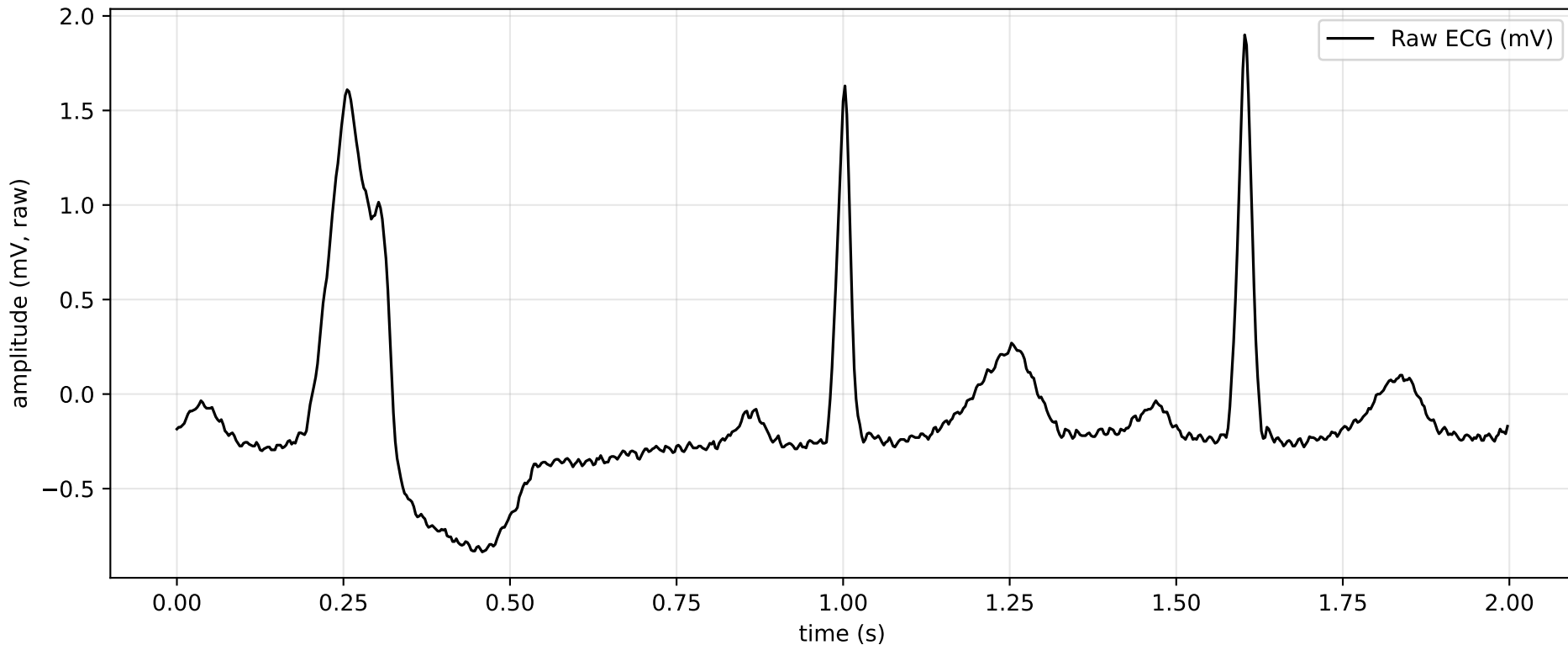
TN: Normal sinus beat (N)



Symbol	N
Type	Normal sinus beat
Source	record 212, sample 365098
Residual	0.0021
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN ✓ correct

NORMAL

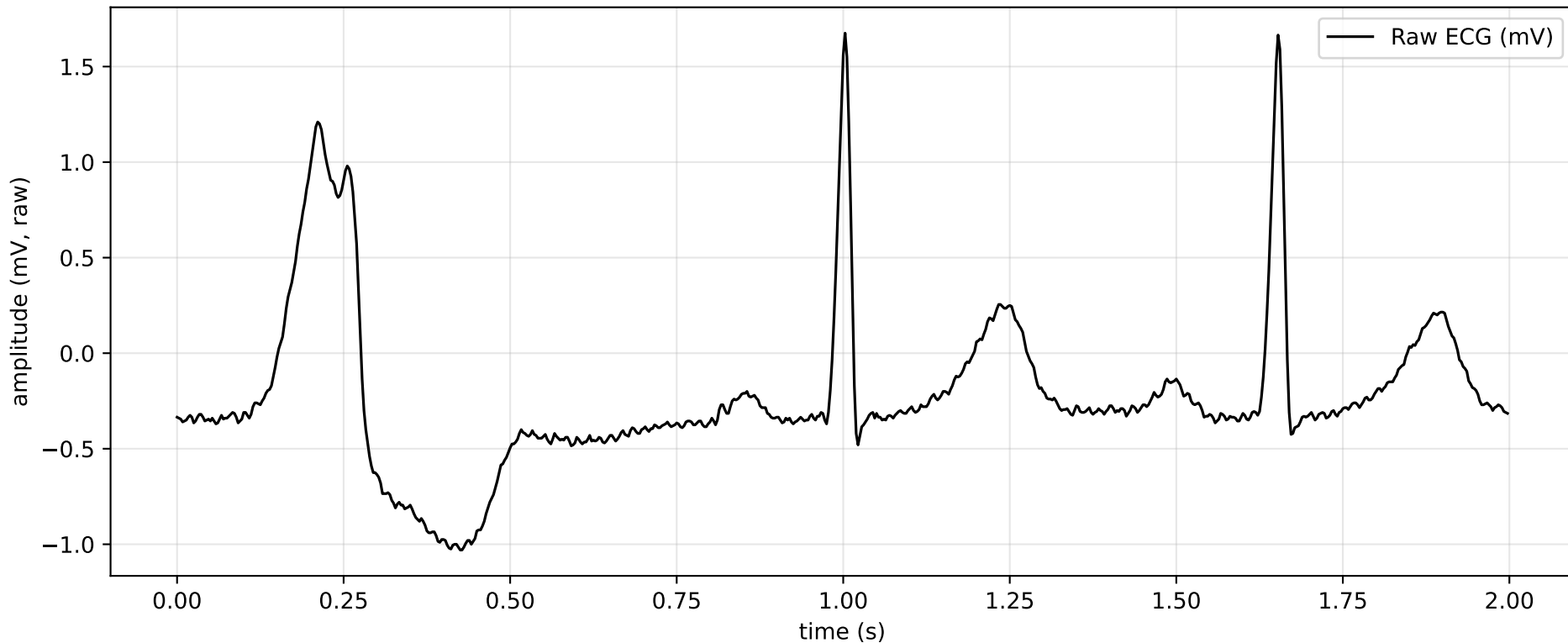
TN: Normal sinus beat (N)



Symbol	N
Type	Normal sinus beat
Source	record 208, sample 249724
Residual	0.0021
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN ✓ correct

NORMAL

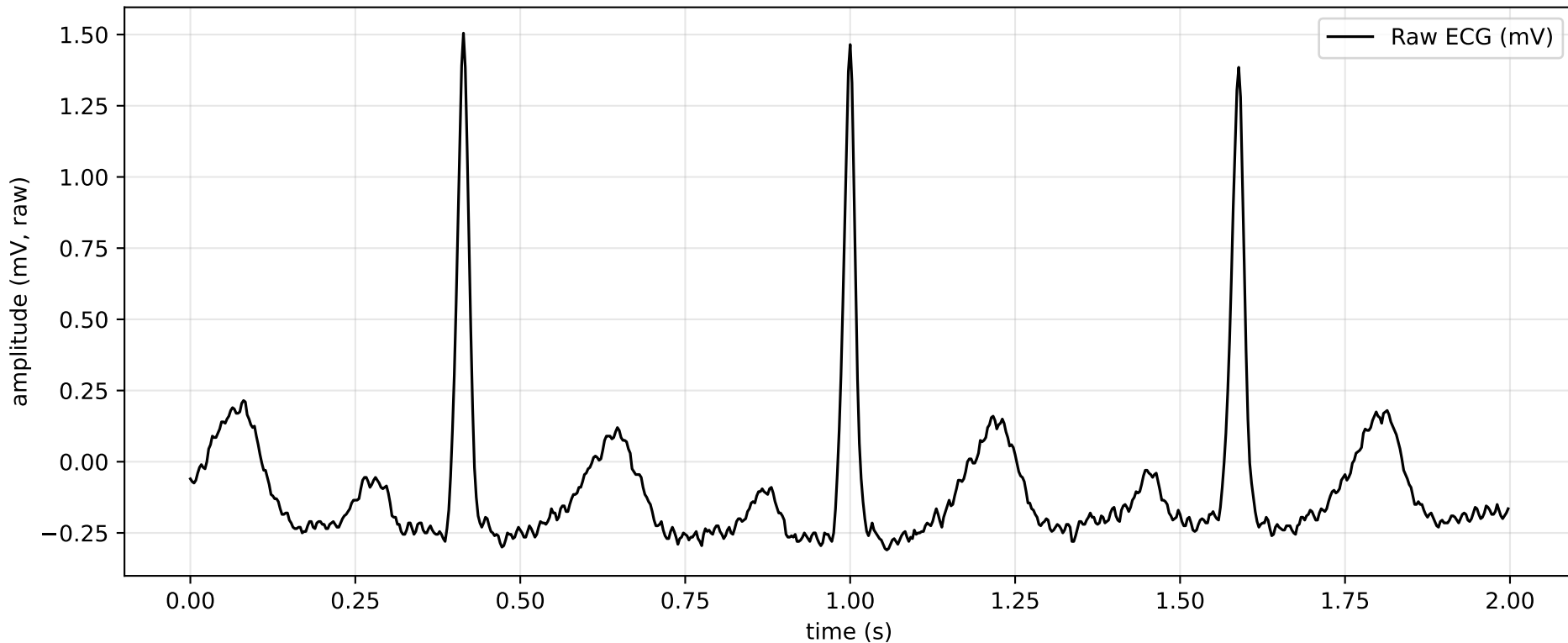
TN: Normal sinus beat (N)



Symbol	N
Type	Normal sinus beat
Source	record 208, sample 578258
Residual	0.0021
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN ✓ correct

NORMAL

TN: Normal sinus beat (N)

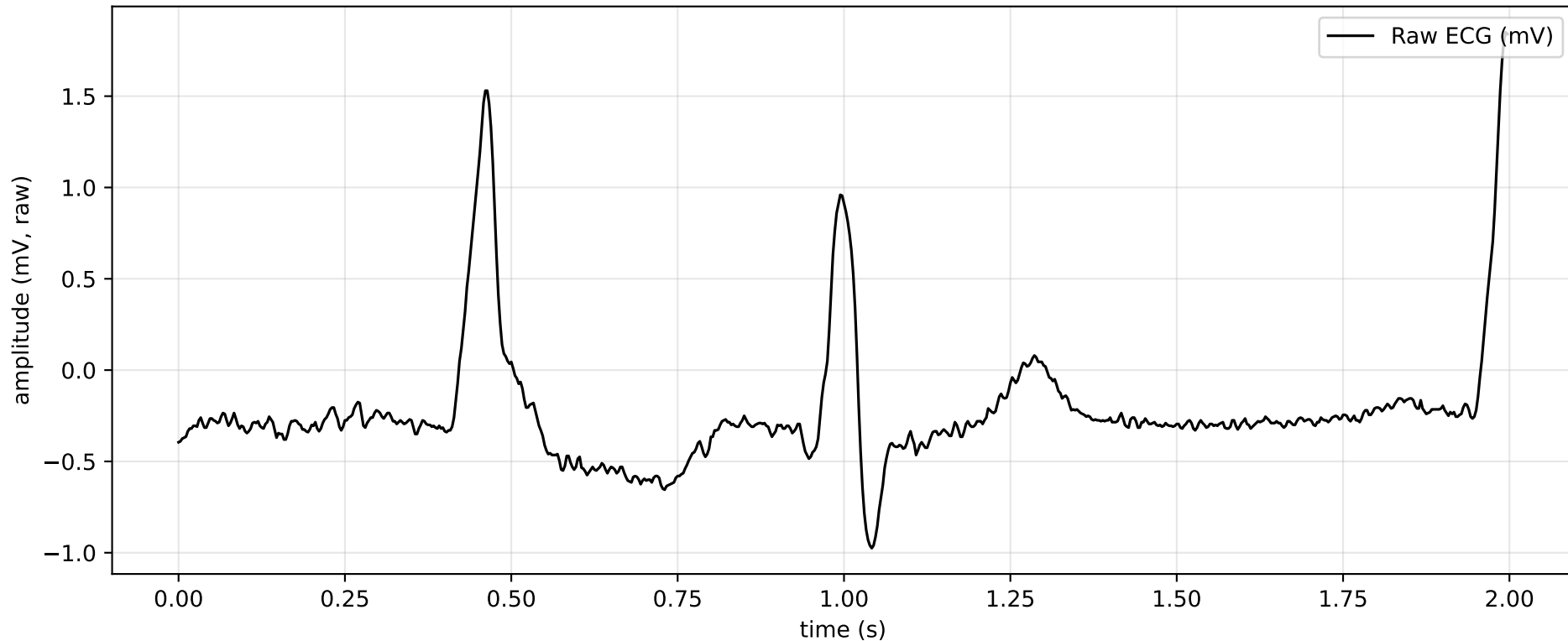


Symbol	N
Type	Normal sinus beat
Source	record 208, sample 107684
Residual	0.0021
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN ✓ correct

NORMAL

Section: V - Premature ventricular contraction (PVC)

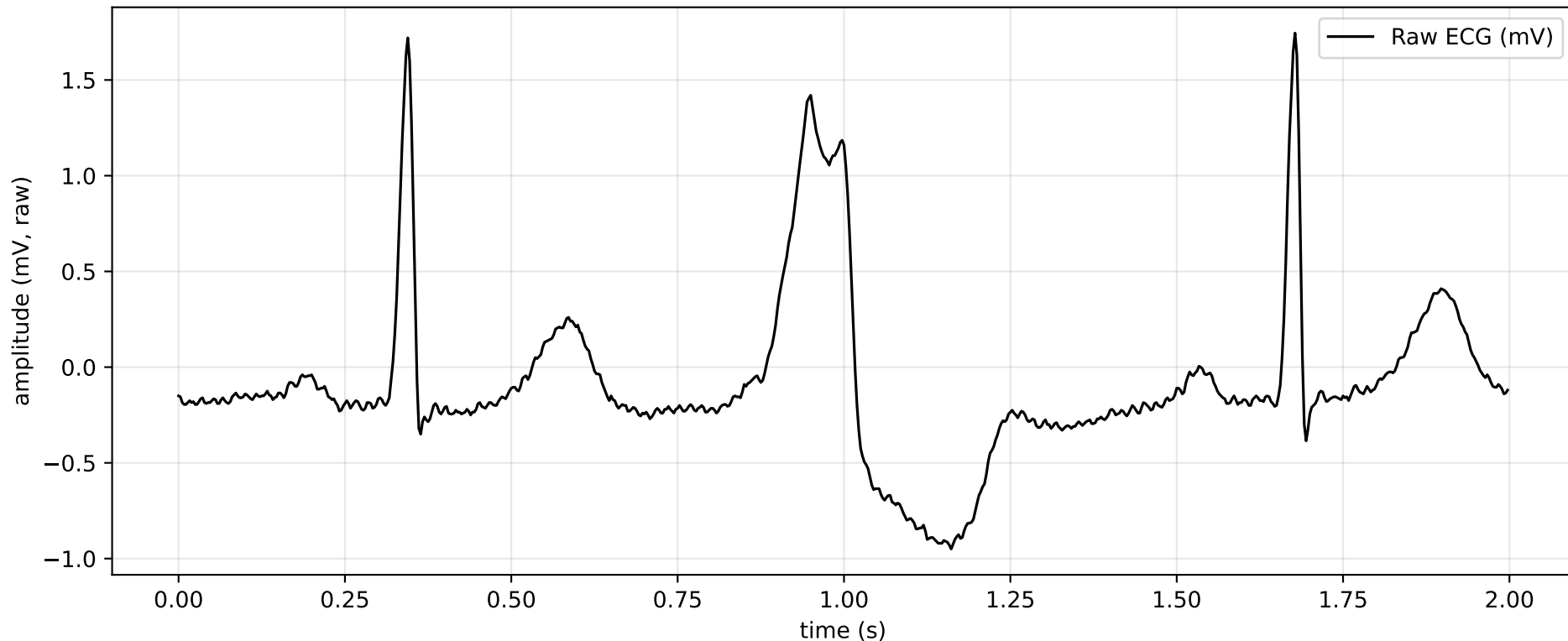
FN: Premature ventricular contraction (PVC) (V)



Symbol	V
Type	Premature ventricular contraction (PVC)
Source	record 214, sample 31486
Residual	0.0427
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

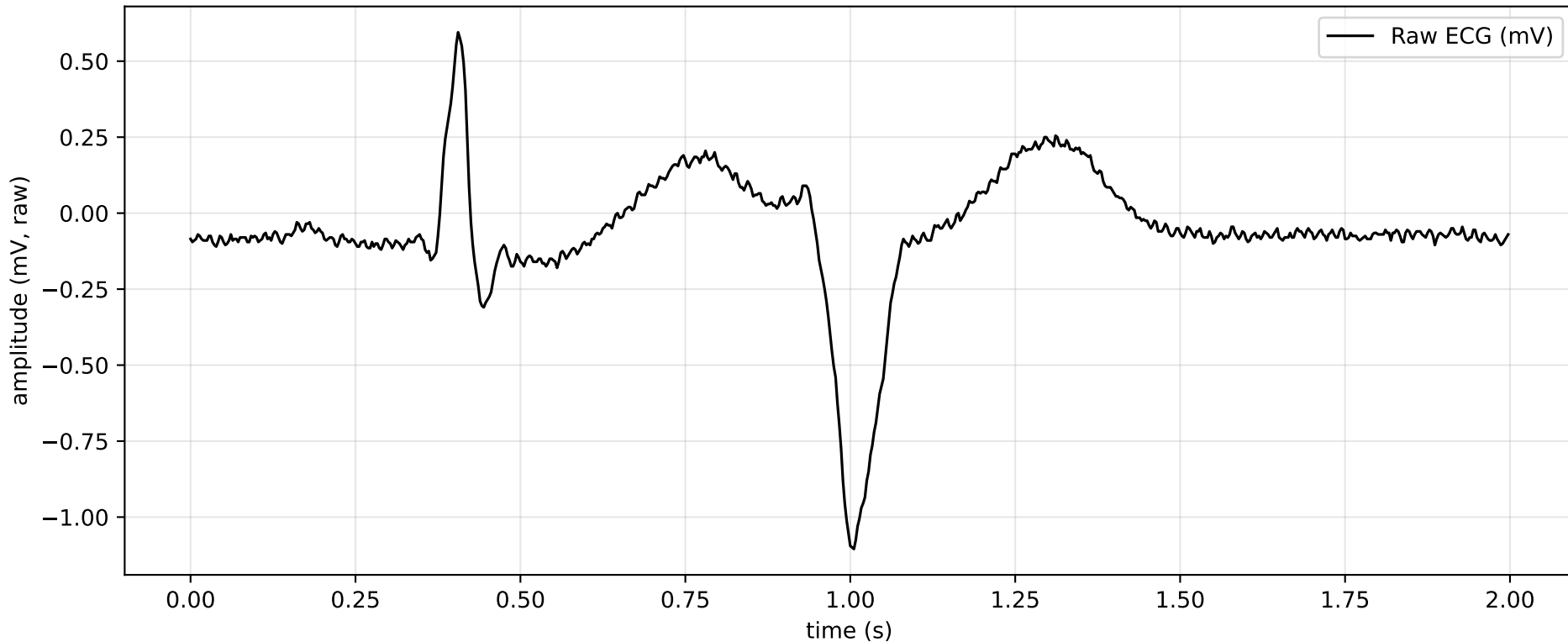
FN: Premature ventricular contraction (PVC) (V)



Symbol	V
Type	Premature ventricular contraction (PVC)
Source	record 208, sample 561101
Residual	0.0423
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

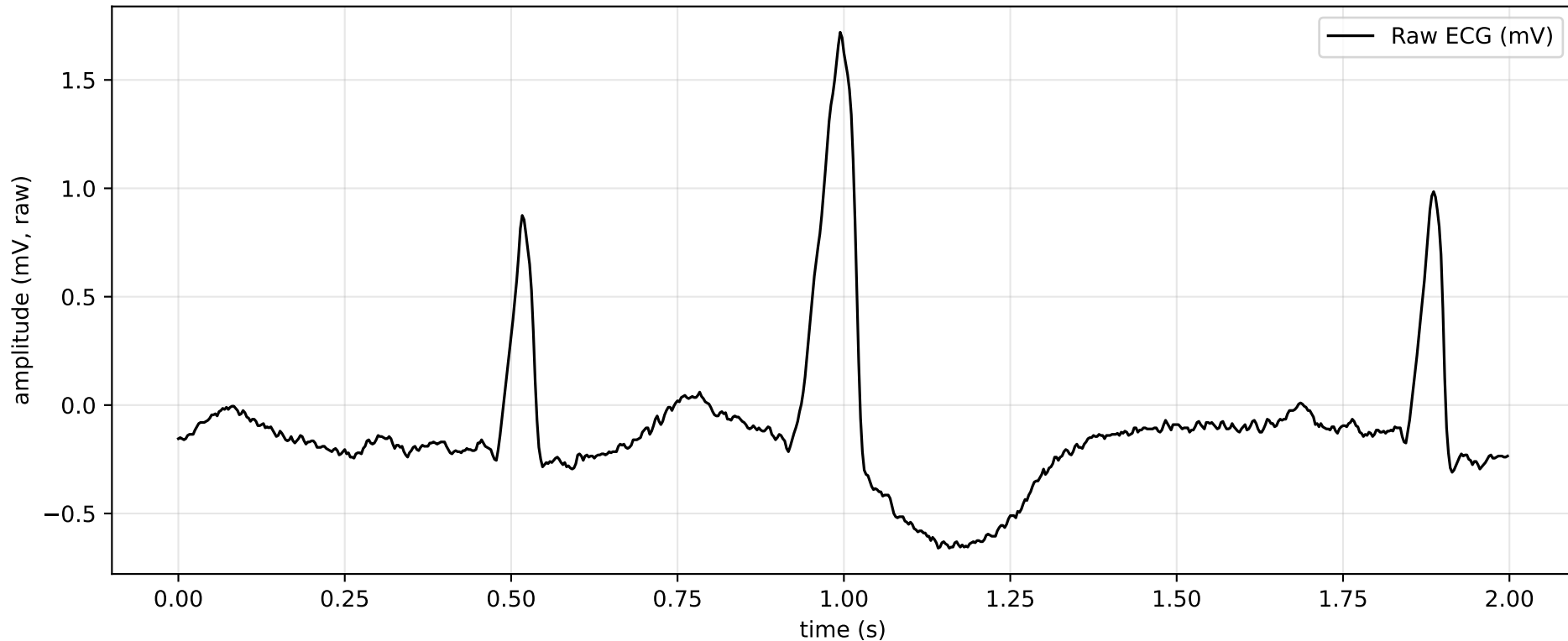
FN: Premature ventricular contraction (PVC) (V)



Symbol	V
Type	Premature ventricular contraction (PVC)
Source	record 207, sample 48883
Residual	0.0422
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

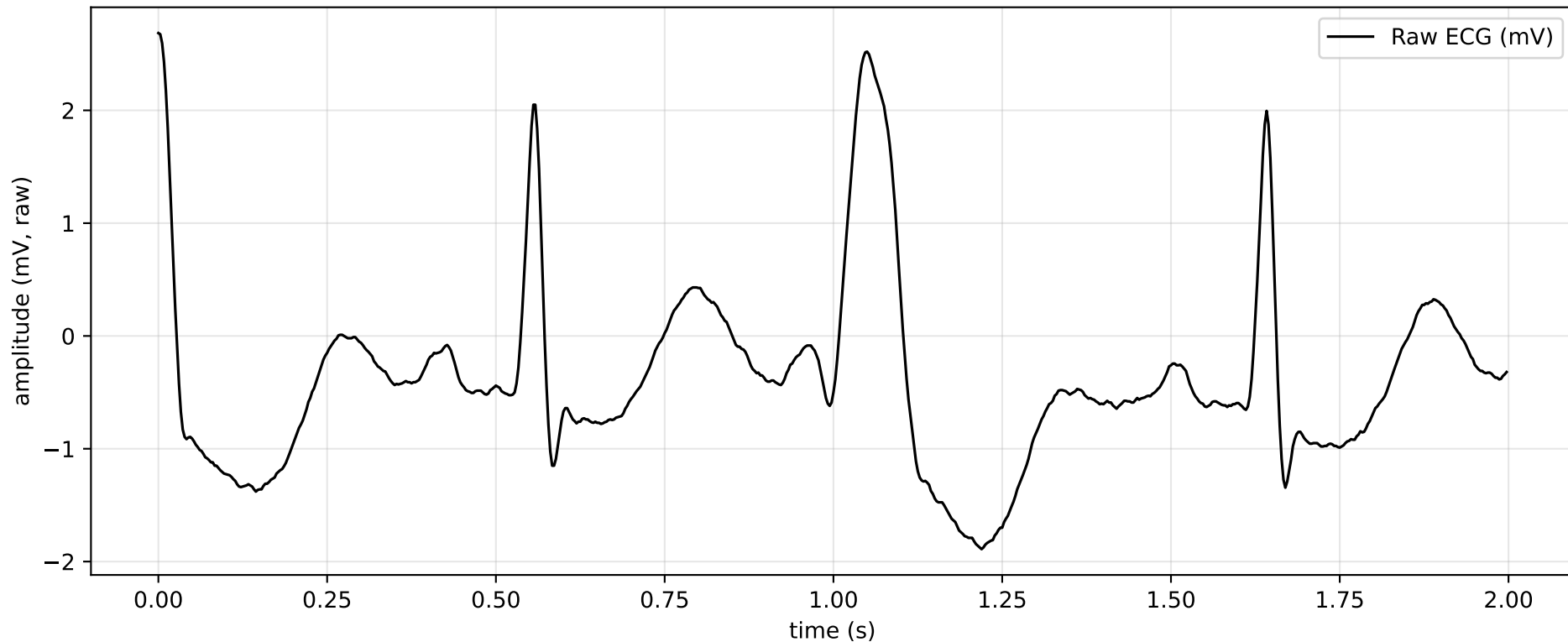
FN: Premature ventricular contraction (PVC) (V)



Symbol	V
Type	Premature ventricular contraction (PVC)
Source	record 200, sample 266894
Residual	0.0420
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

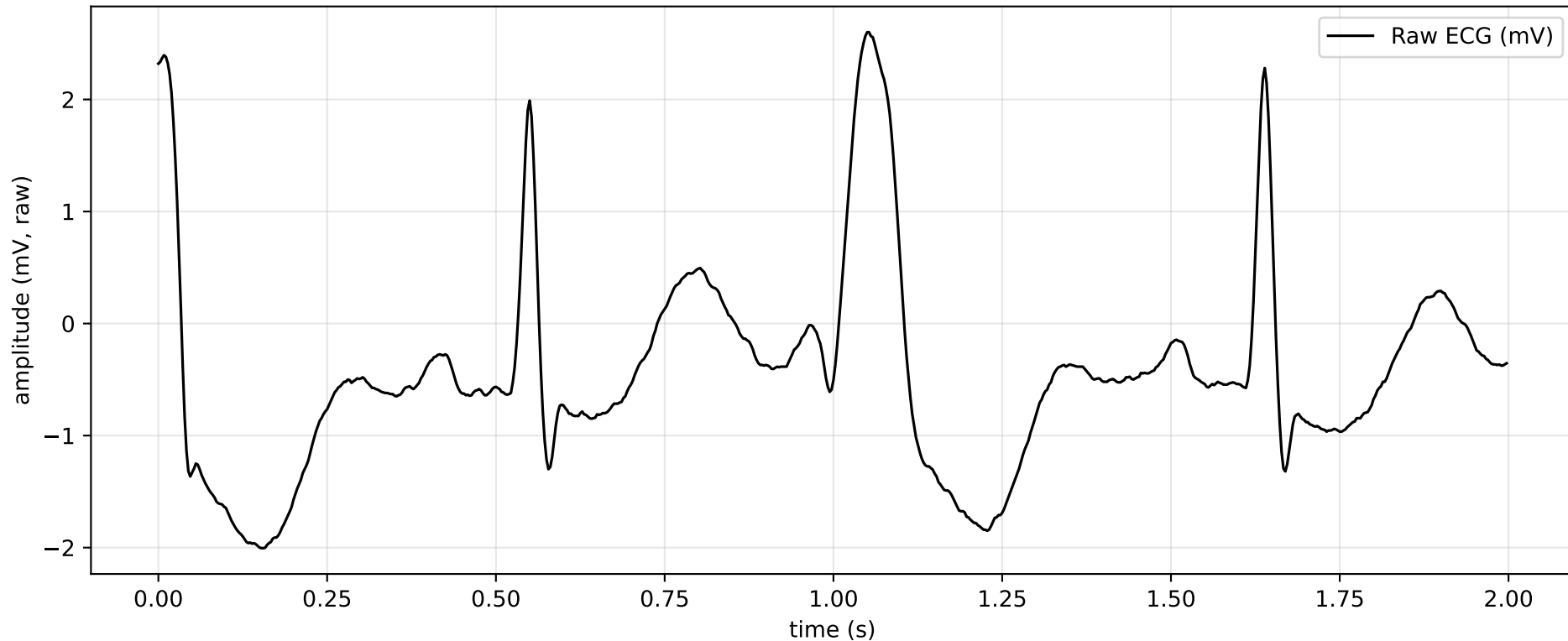
TP: Premature ventricular contraction (PVC) (V)



Symbol	V
Type	Premature ventricular contraction (PVC)
Source	record 213, sample 57428
Residual	1.2749
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

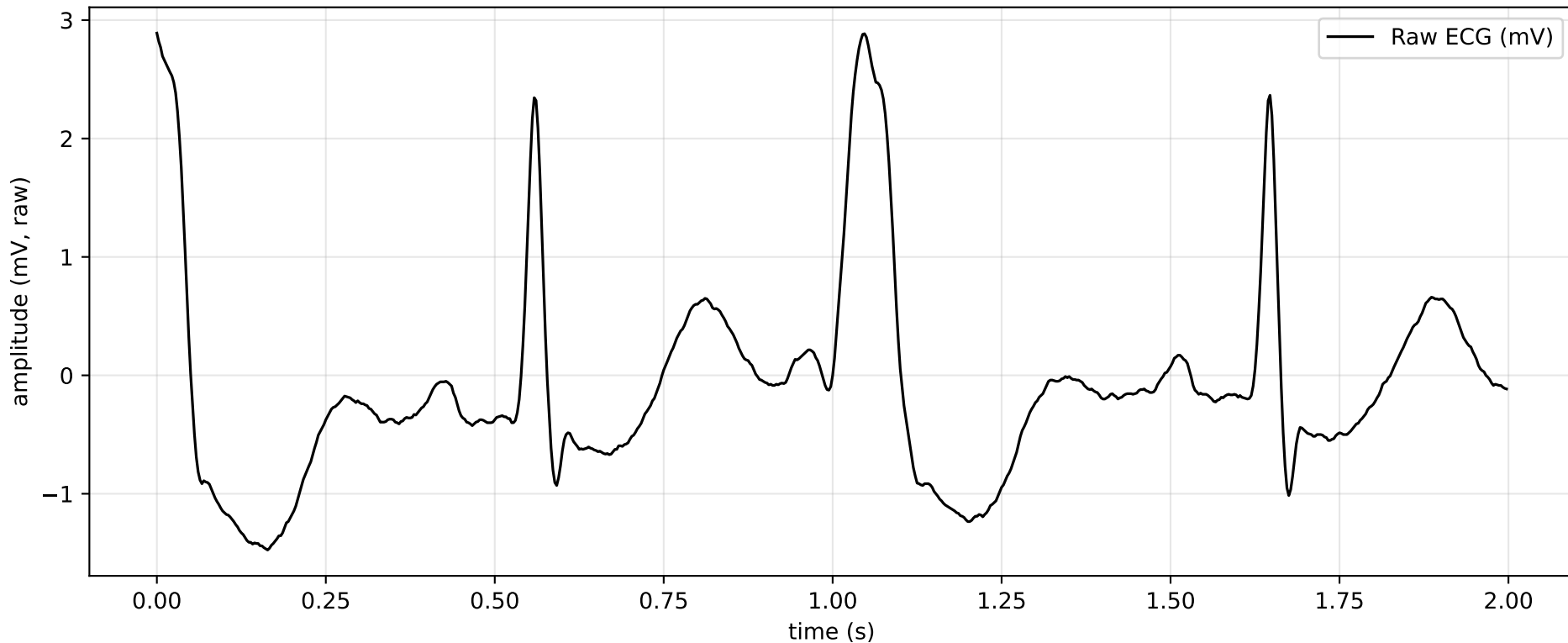
TP: Premature ventricular contraction (PVC) (V)



Symbol	V
Type	Premature ventricular contraction (PVC)
Source	record 213, sample 96670
Residual	1.2723
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

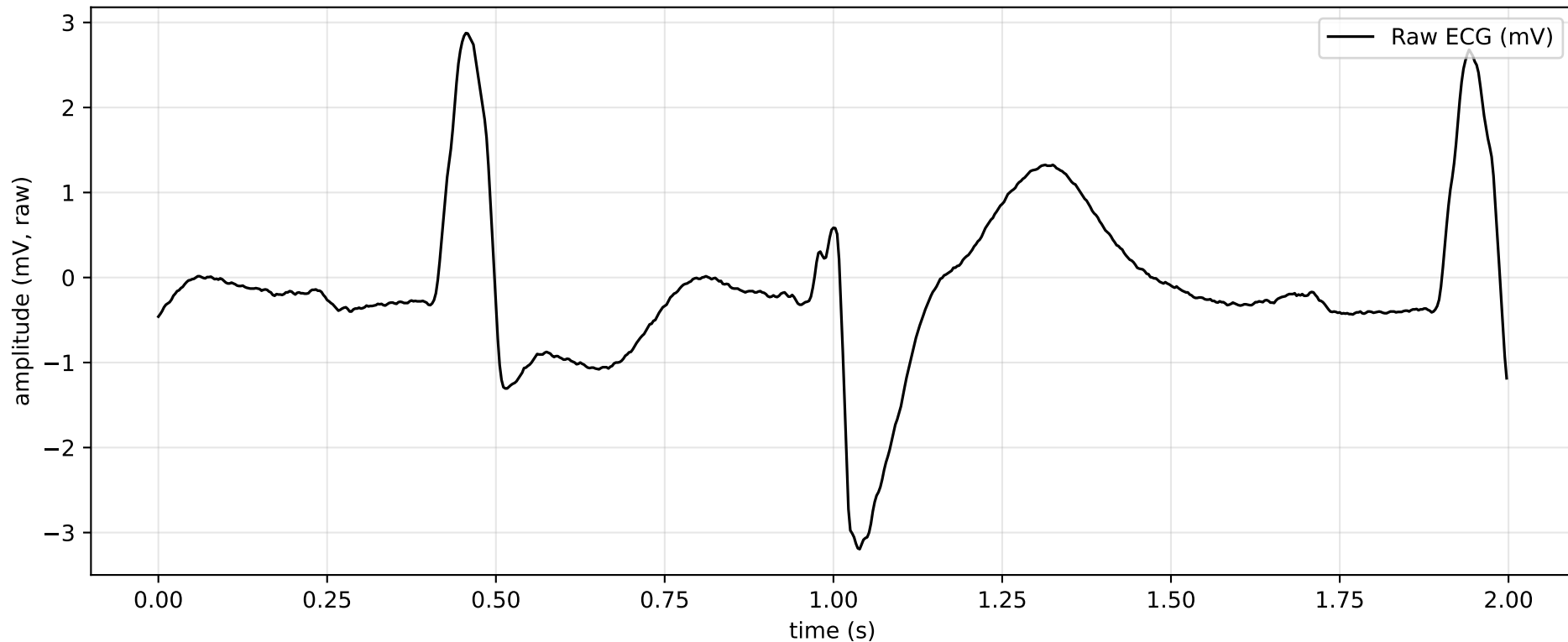
TP: Premature ventricular contraction (PVC) (V)



Symbol	V
Type	Premature ventricular contraction (PVC)
Source	record 213, sample 167531
Residual	1.2521
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

TP: Premature ventricular contraction (PVC) (V)

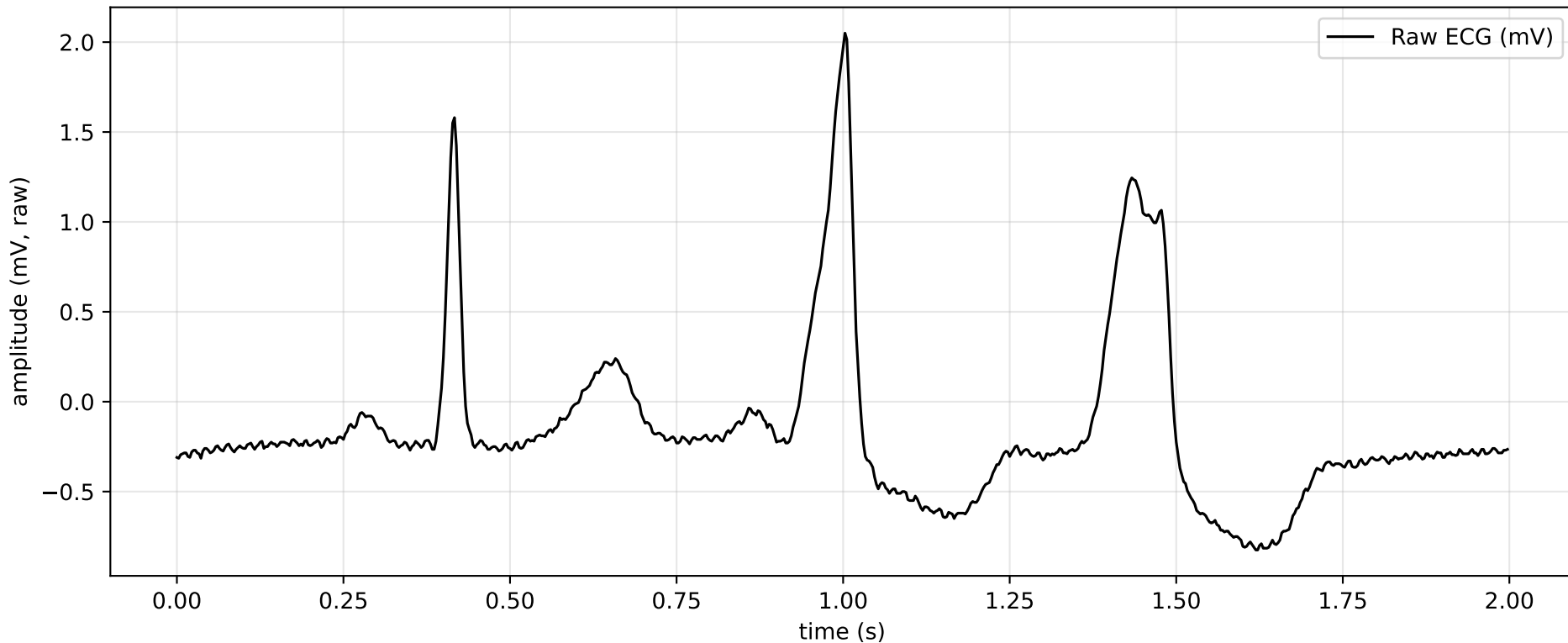


Symbol	V
Type	Premature ventricular contraction (PVC)
Source	record 109, sample 603535
Residual	1.1877
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

Section: F - Fusion of ventricular and normal

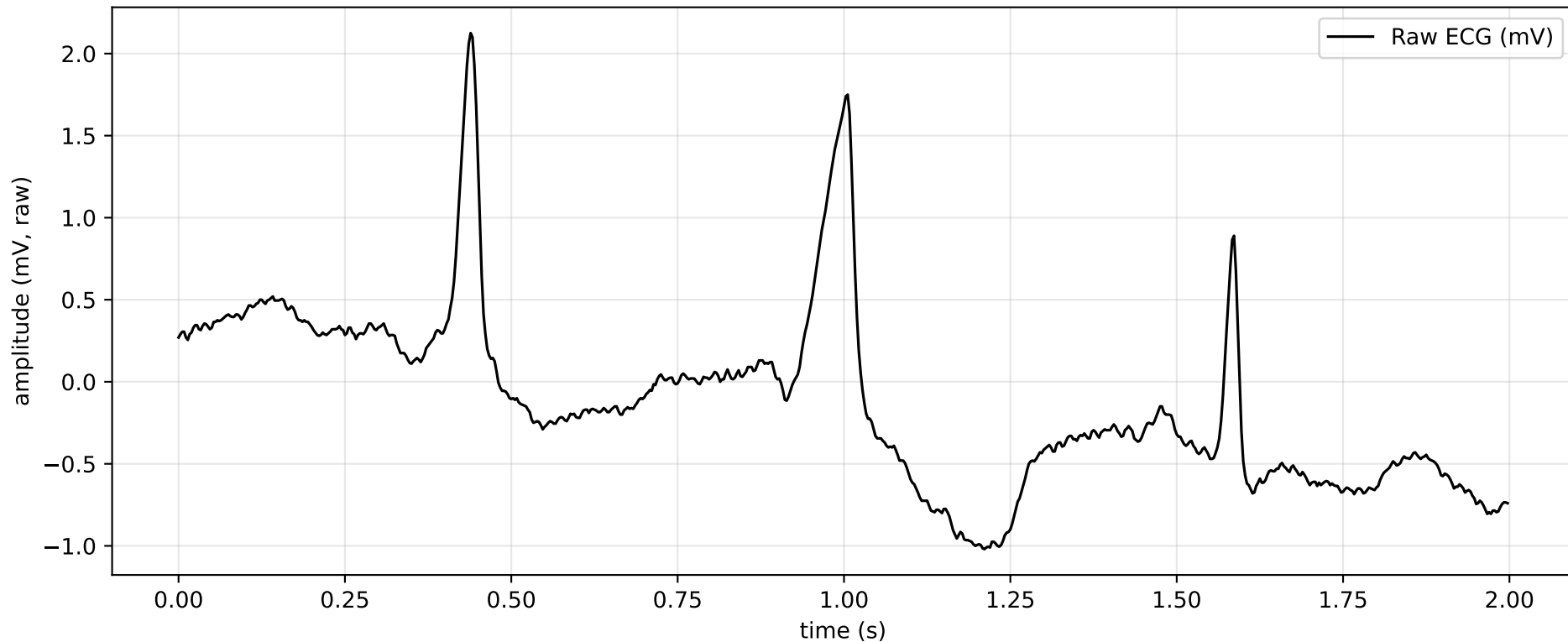
FN: Fusion of ventricular and normal (F)



Symbol	F
Type	Fusion of ventricular and normal
Source	record 208, sample 203236
Residual	0.0434
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

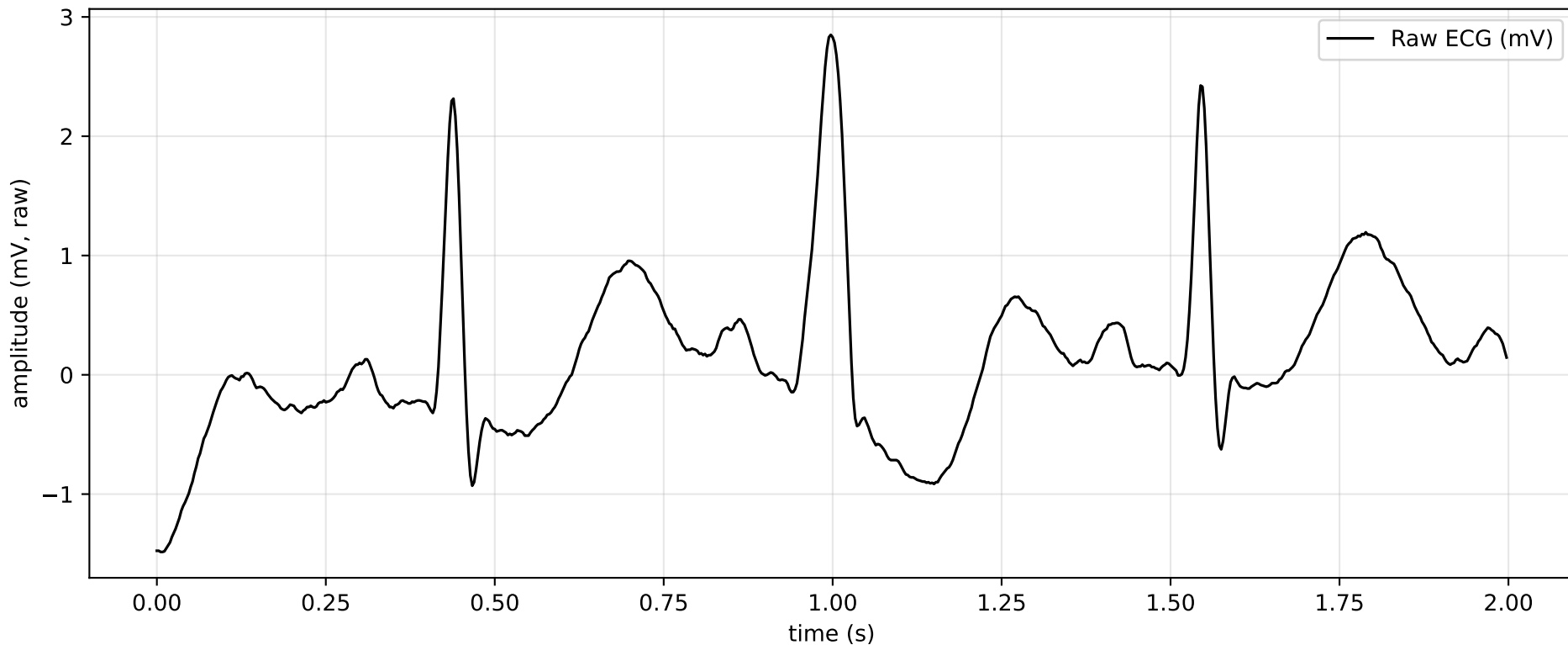
FN: Fusion of ventricular and normal (F)



Symbol	F
Type	Fusion of ventricular and normal
Source	record 208, sample 429272
Residual	0.0434
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

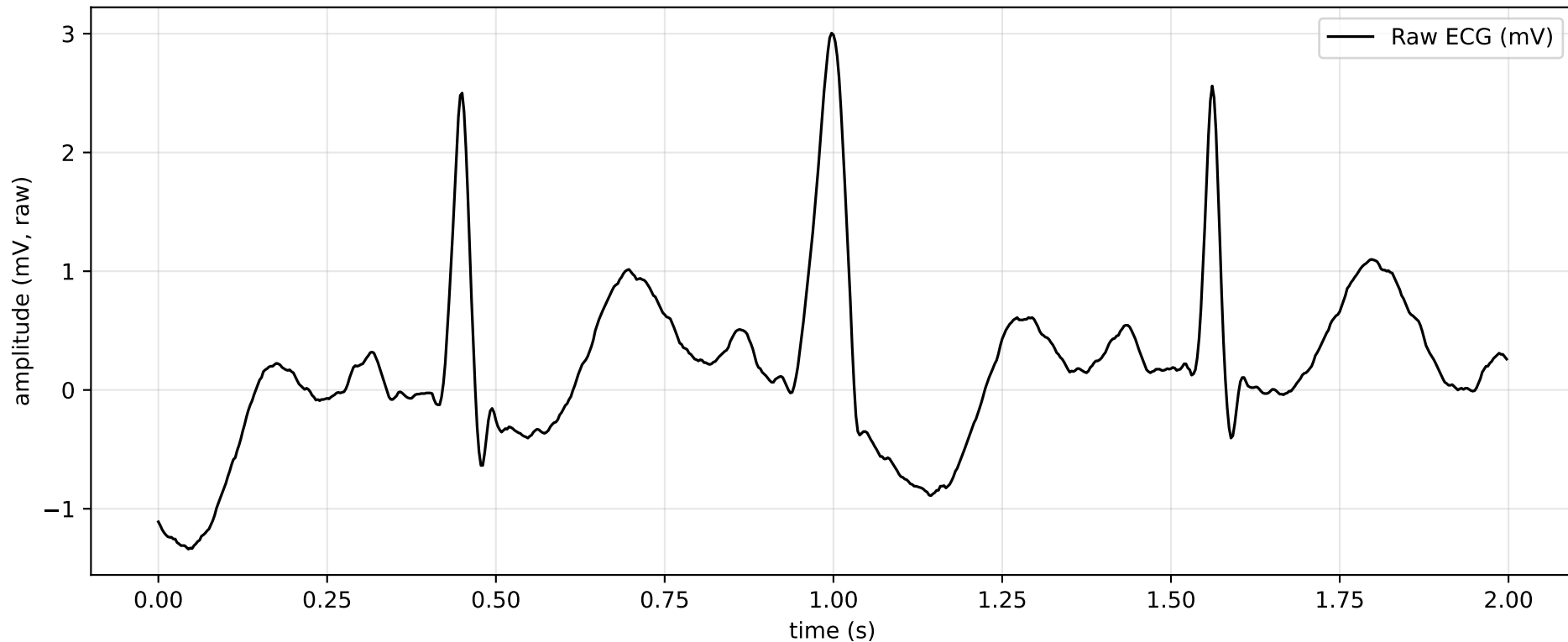
FN: Fusion of ventricular and normal (F)



Symbol	F
Type	Fusion of ventricular and normal
Source	record 213, sample 377920
Residual	0.0433
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

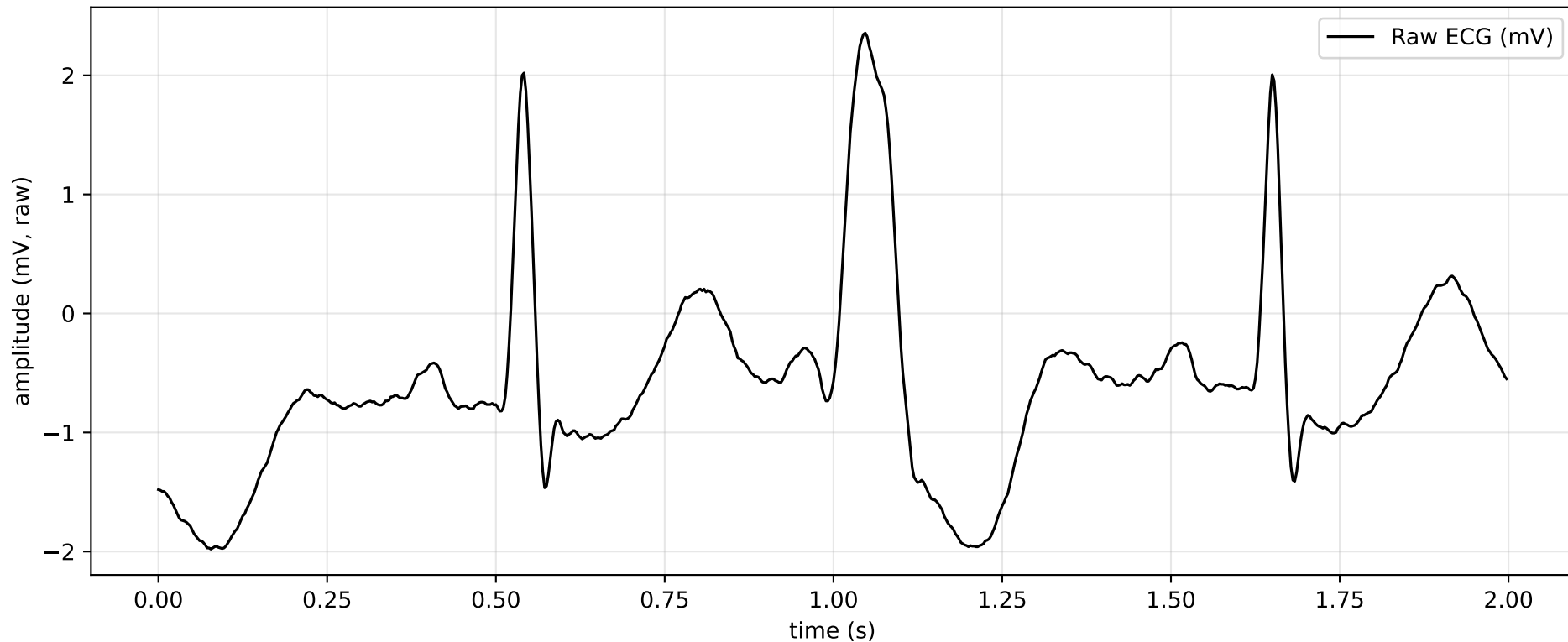
FN: Fusion of ventricular and normal (F)



Symbol	F
Type	Fusion of ventricular and normal
Source	record 213, sample 406607
Residual	0.0431
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

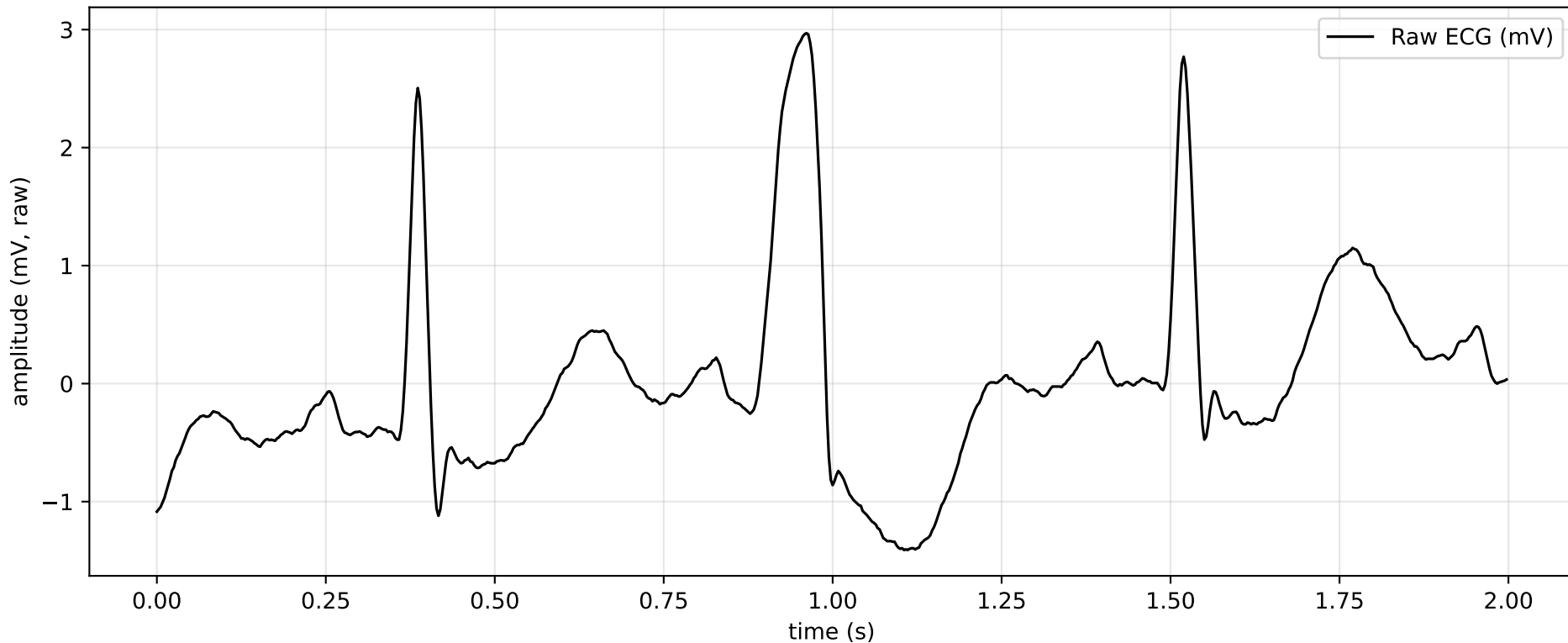
TP: Fusion of ventricular and normal (F)



Symbol	F
Type	Fusion of ventricular and normal
Source	record 213, sample 380292
Residual	0.6744
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

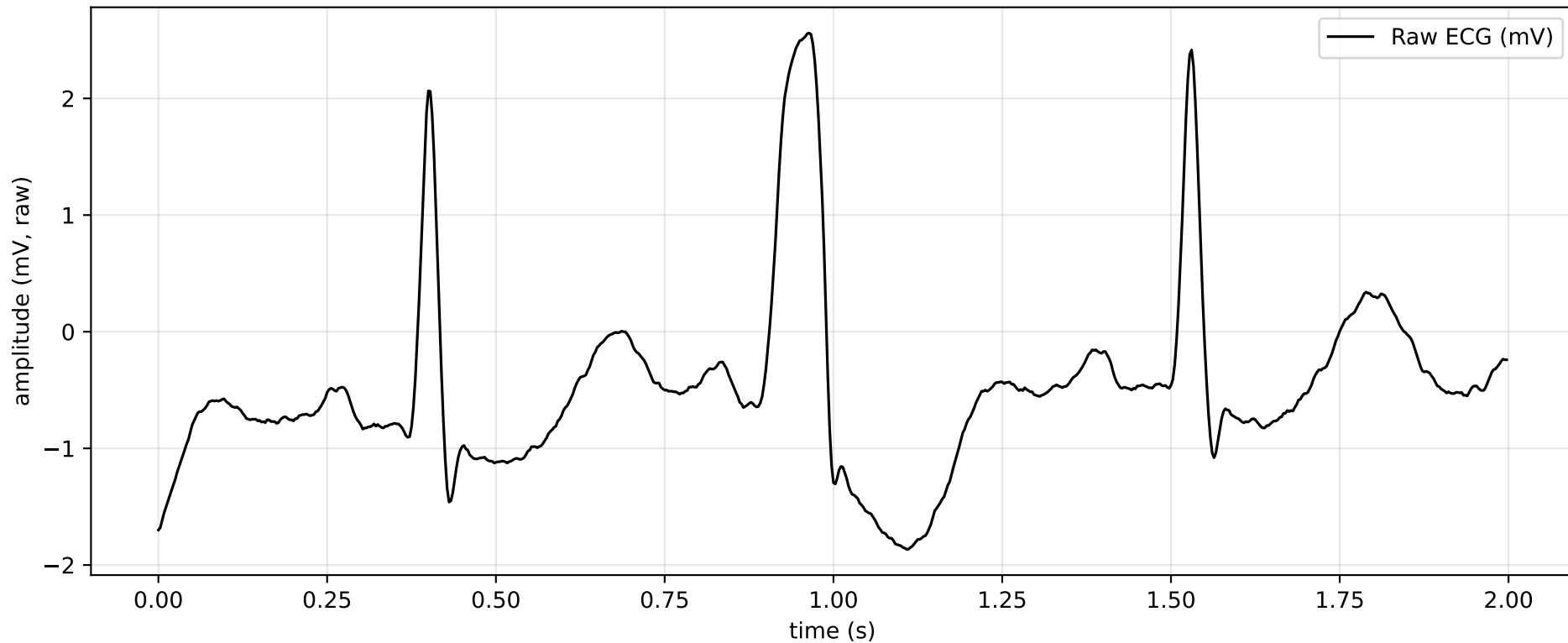
TP: Fusion of ventricular and normal (F)



Symbol	F
Type	Fusion of ventricular and normal
Source	record 213, sample 477982
Residual	0.2730
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

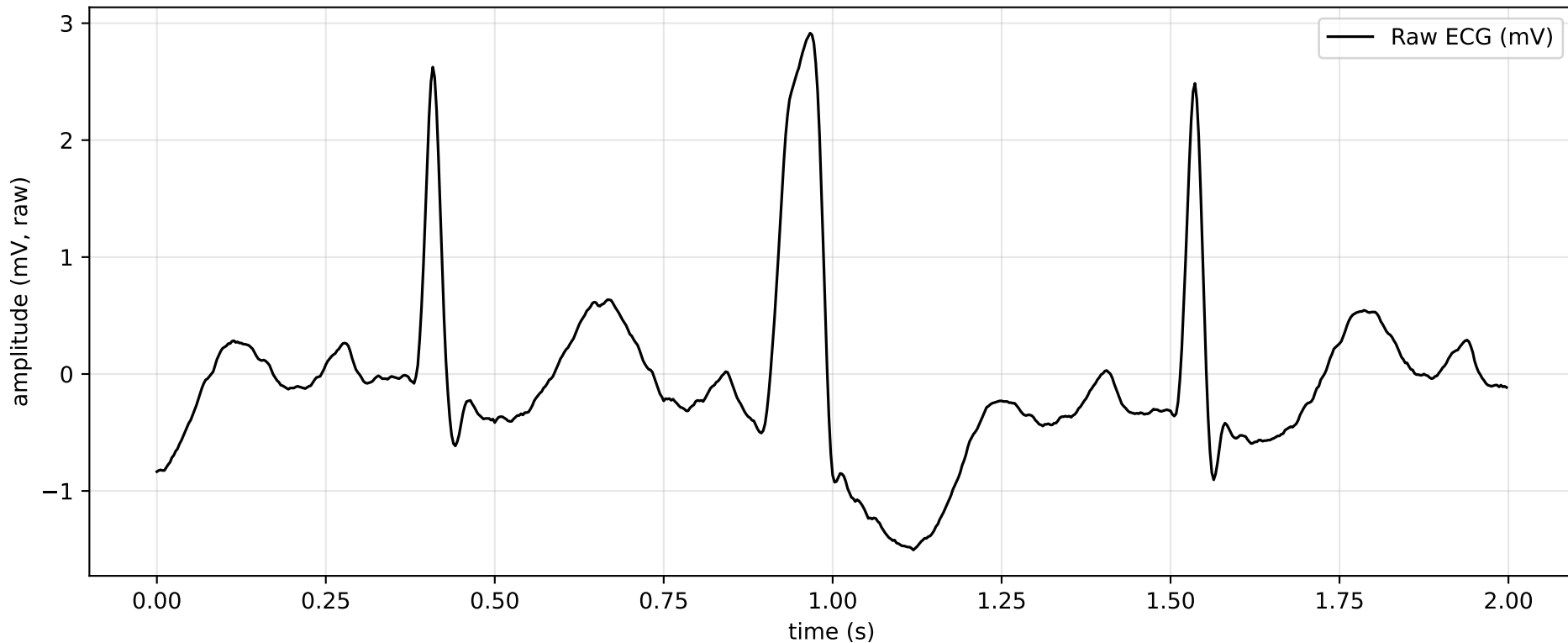
TP: Fusion of ventricular and normal (F)



Symbol	F
Type	Fusion of ventricular and normal
Source	record 213, sample 483675
Residual	0.2259
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

TP: Fusion of ventricular and normal (F)

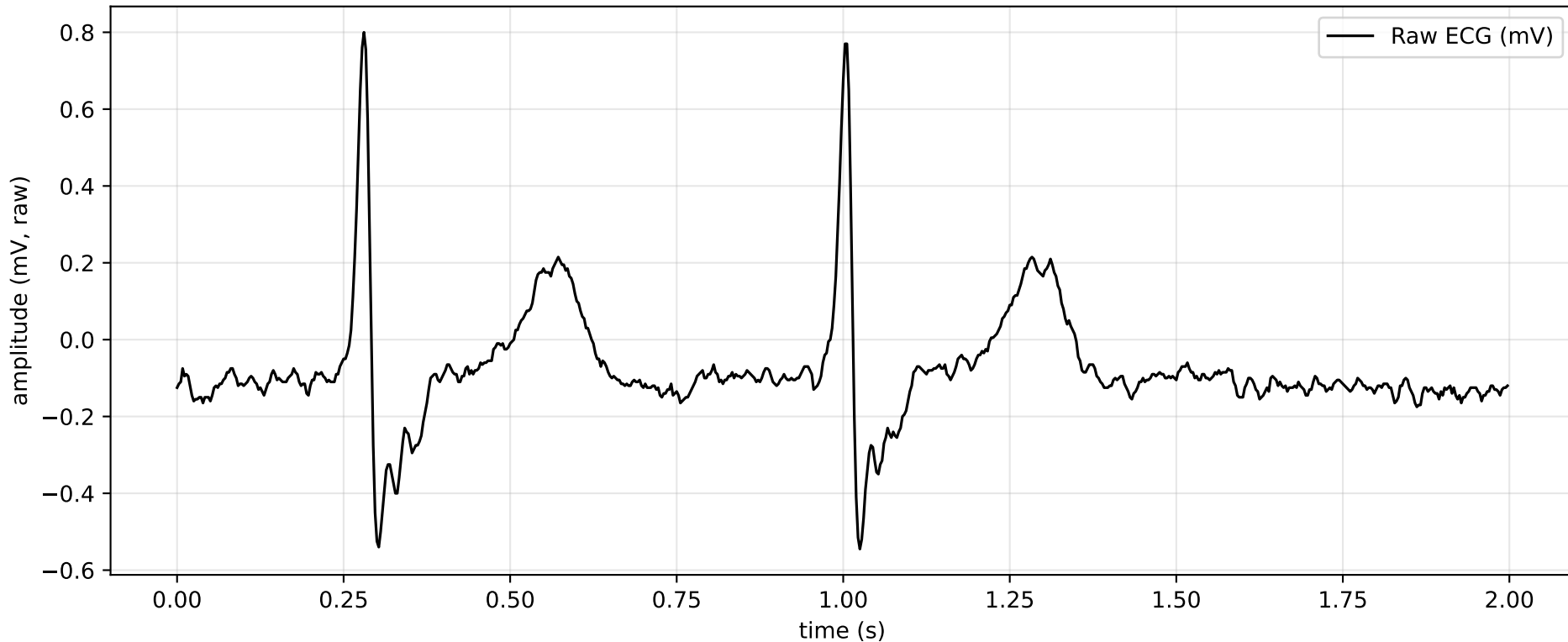


Symbol	F
Type	Fusion of ventricular and normal
Source	record 213, sample 485711
Residual	0.2060
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

Section: A - Atrial premature beat (APC)

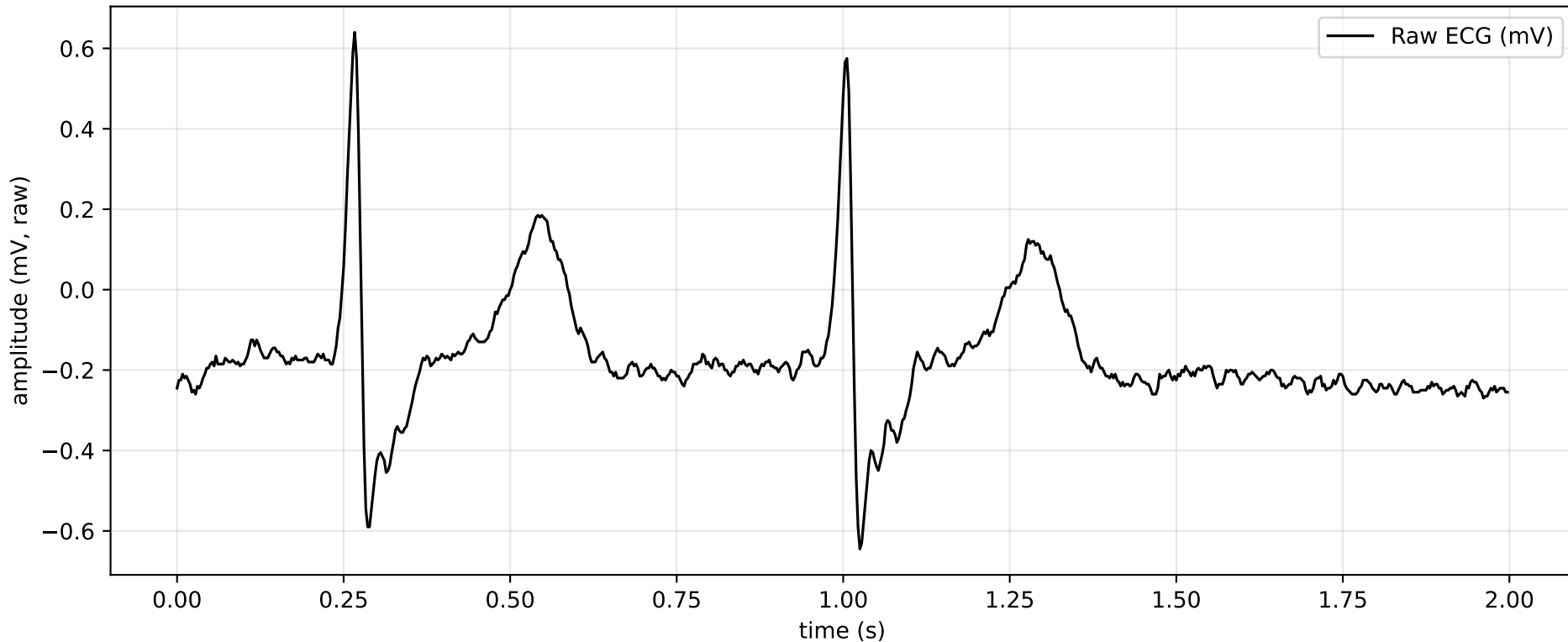
FN: Atrial premature beat (APC) (A)



Symbol	A
Type	Atrial premature beat (APC)
Source	record 232, sample 375665
Residual	0.0434
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

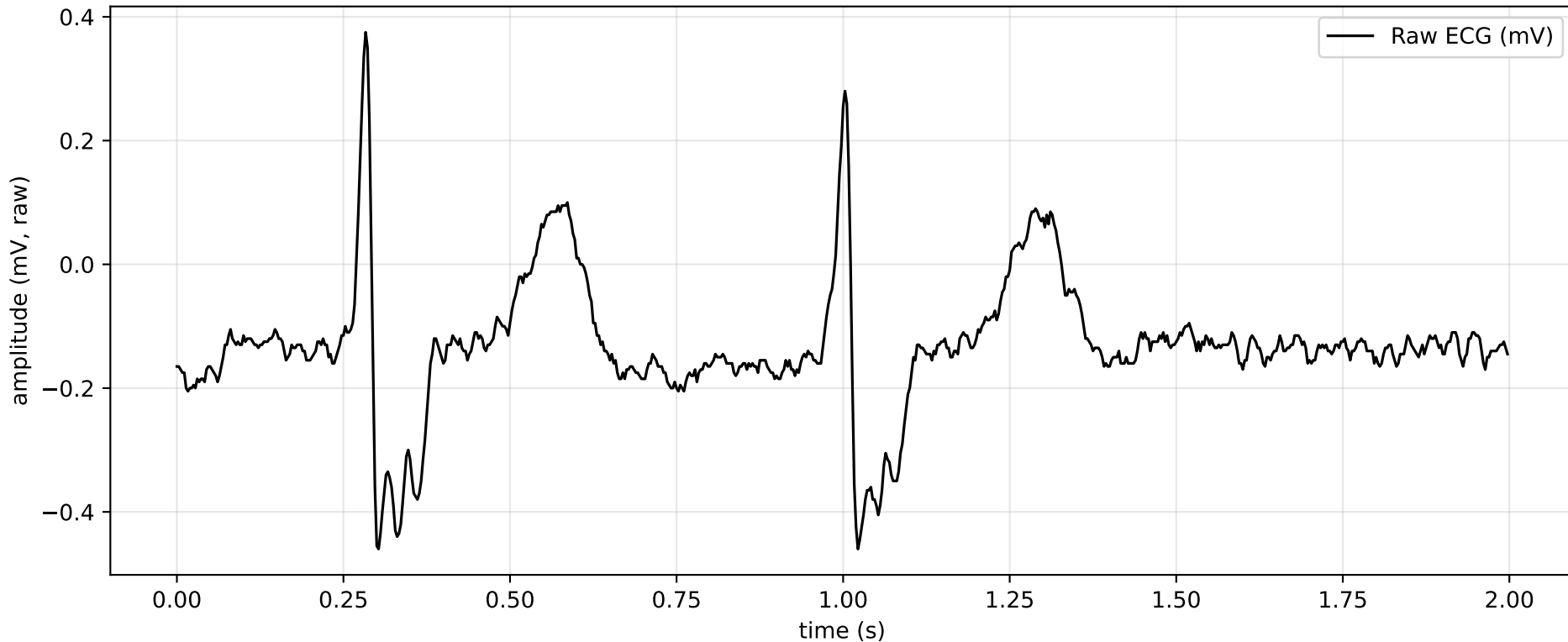
FN: Atrial premature beat (APC) (A)



Symbol	A
Type	Atrial premature beat (APC)
Source	record 232, sample 285562
Residual	0.0433
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

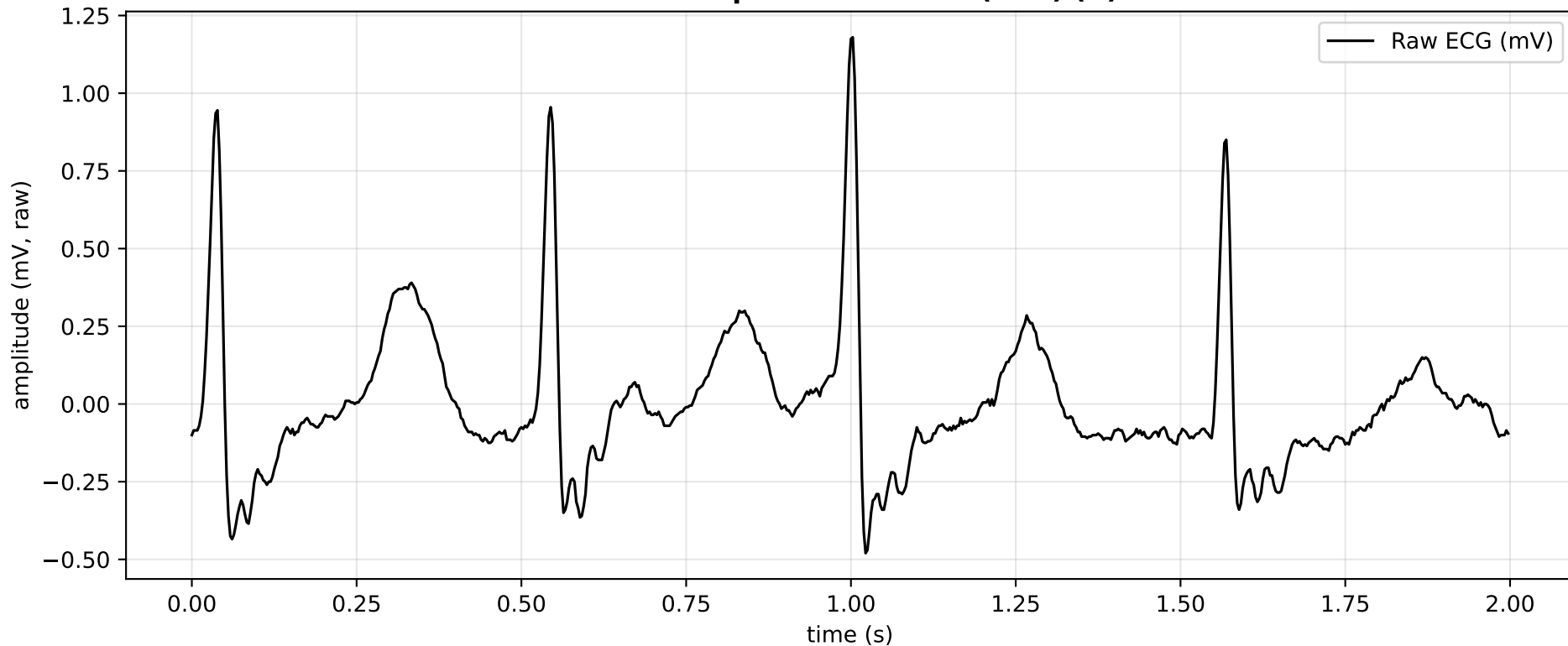
FN: Atrial premature beat (APC) (A)



Symbol	A
Type	Atrial premature beat (APC)
Source	record 232, sample 40168
Residual	0.0433
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

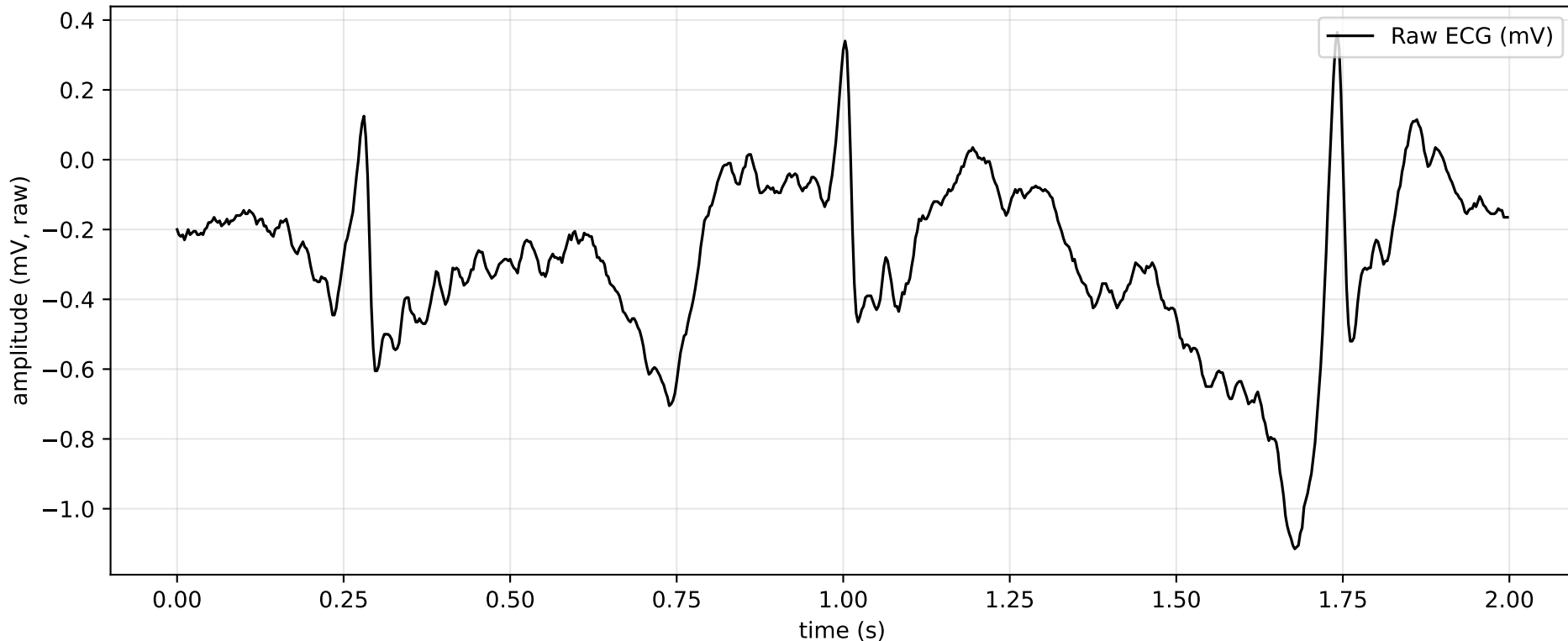
TP: Atrial premature beat (APC) (A)



Symbol	A
Type	Atrial premature beat (APC)
Source	record 232, sample 446678
Residual	1.2566
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

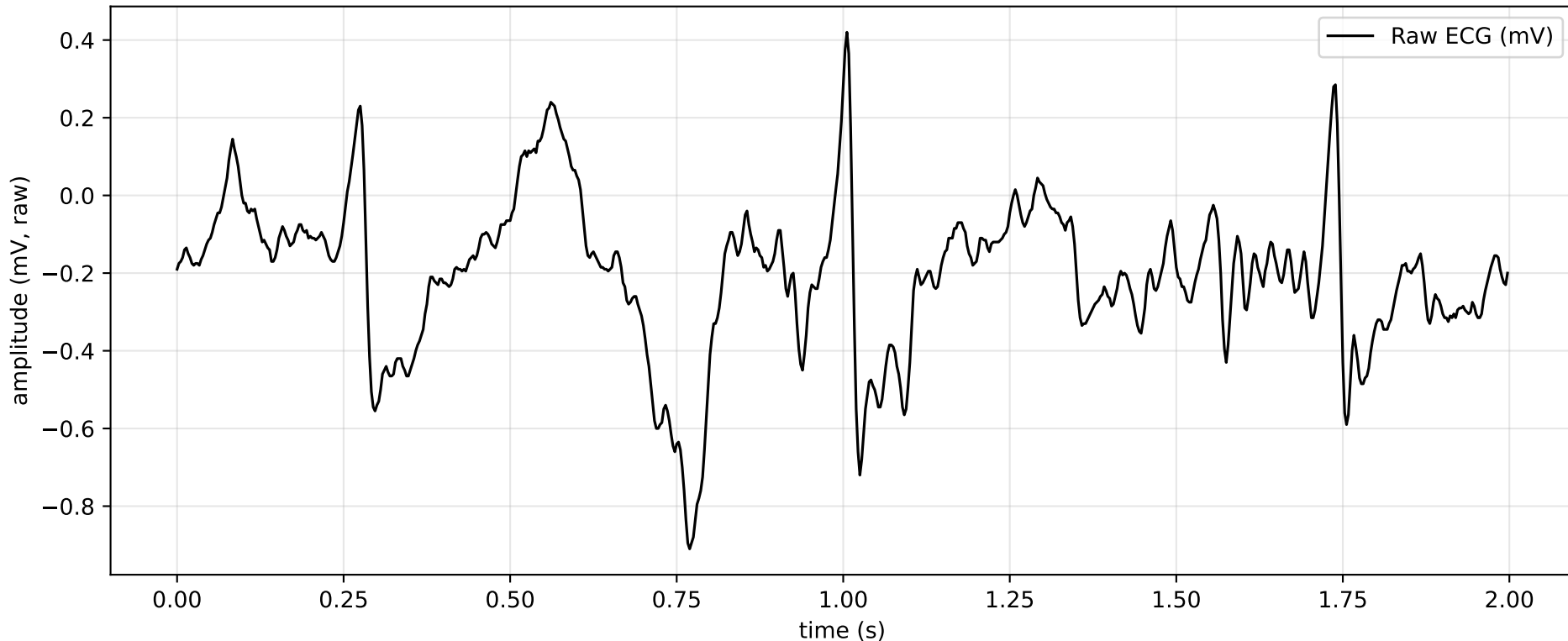
TP: Atrial premature beat (APC) (A)



Symbol	A
Type	Atrial premature beat (APC)
Source	record 232, sample 245833
Residual	1.2324
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

TP: Atrial premature beat (APC) (A)

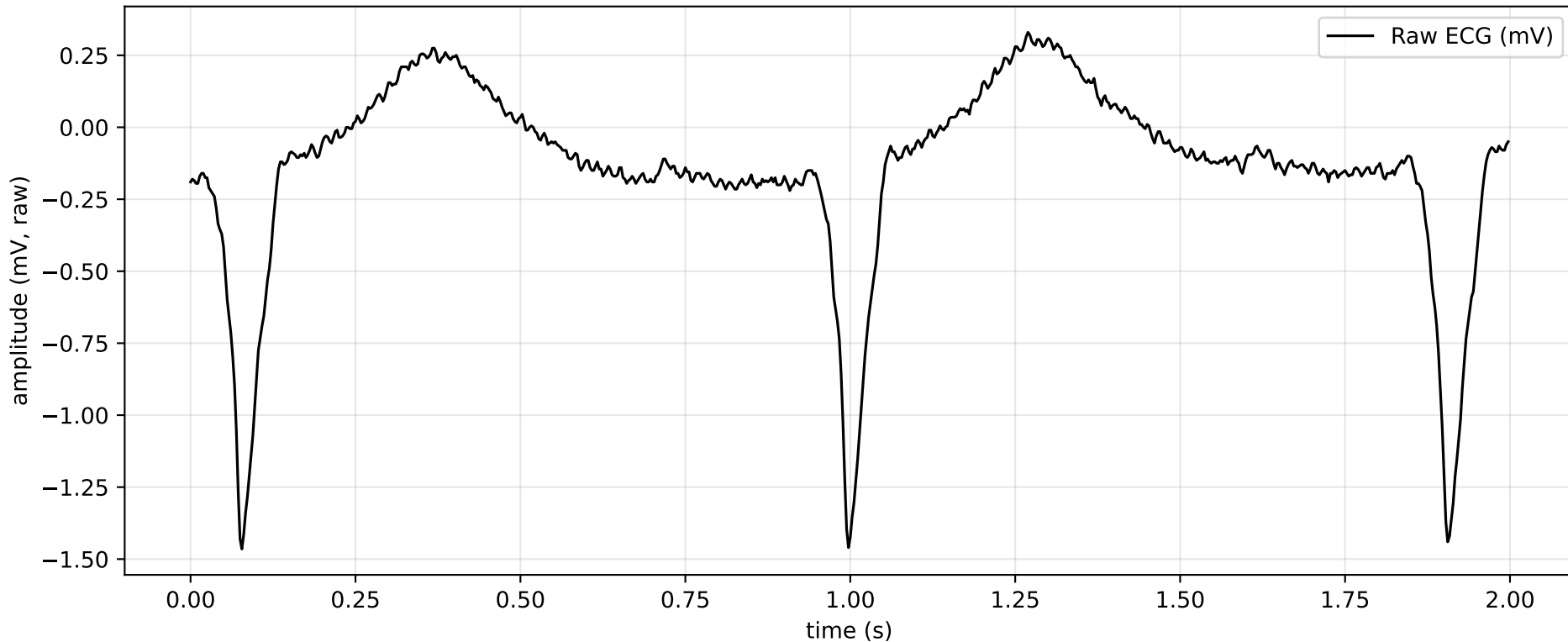


Symbol	A
Type	Atrial premature beat (APC)
Source	record 232, sample 247683
Residual	0.9413
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

Section: L - Left bundle branch block

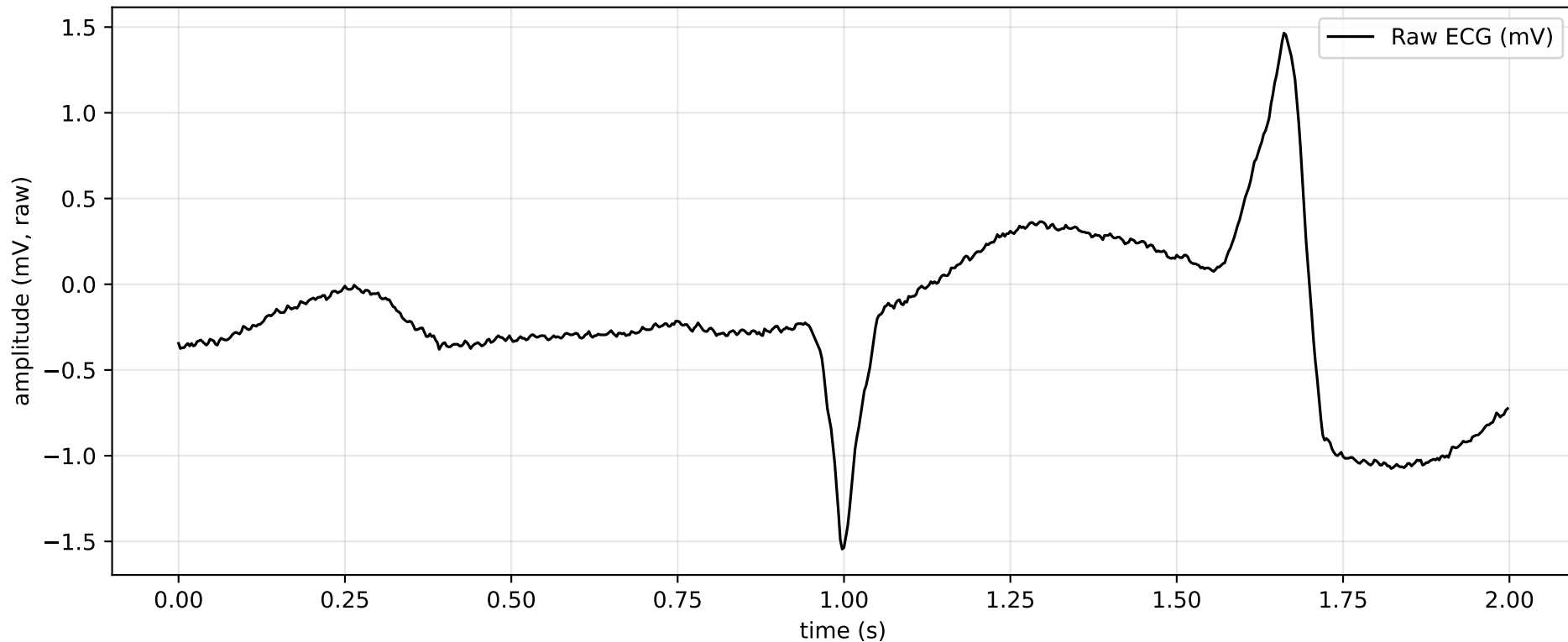
FN: Left bundle branch block (L)



Symbol	L
Type	Left bundle branch block
Source	record 207, sample 288750
Residual	0.0433
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	FN ✓ correct

NORMAL

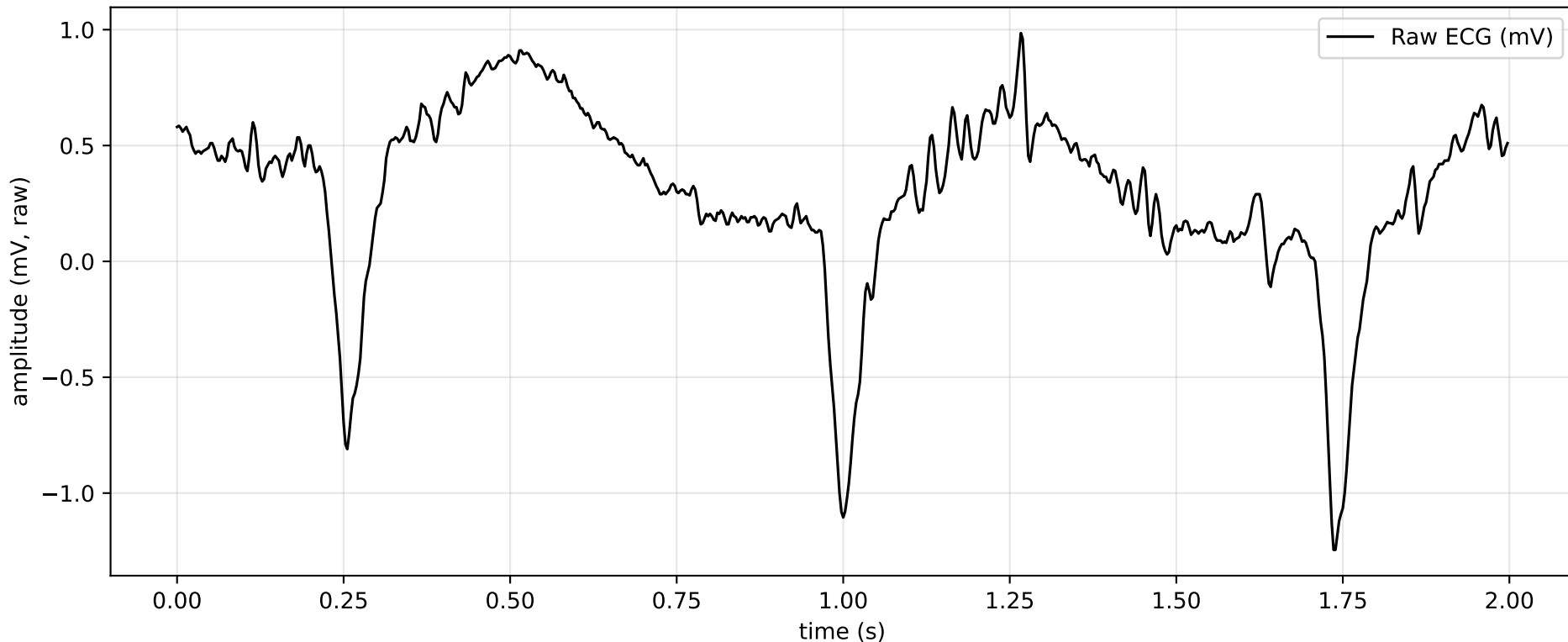
FN: Left bundle branch block (L)



Symbol	L
Type	Left bundle branch block
Source	record 207, sample 29847
Residual	0.0433
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	FN ✓ correct

NORMAL

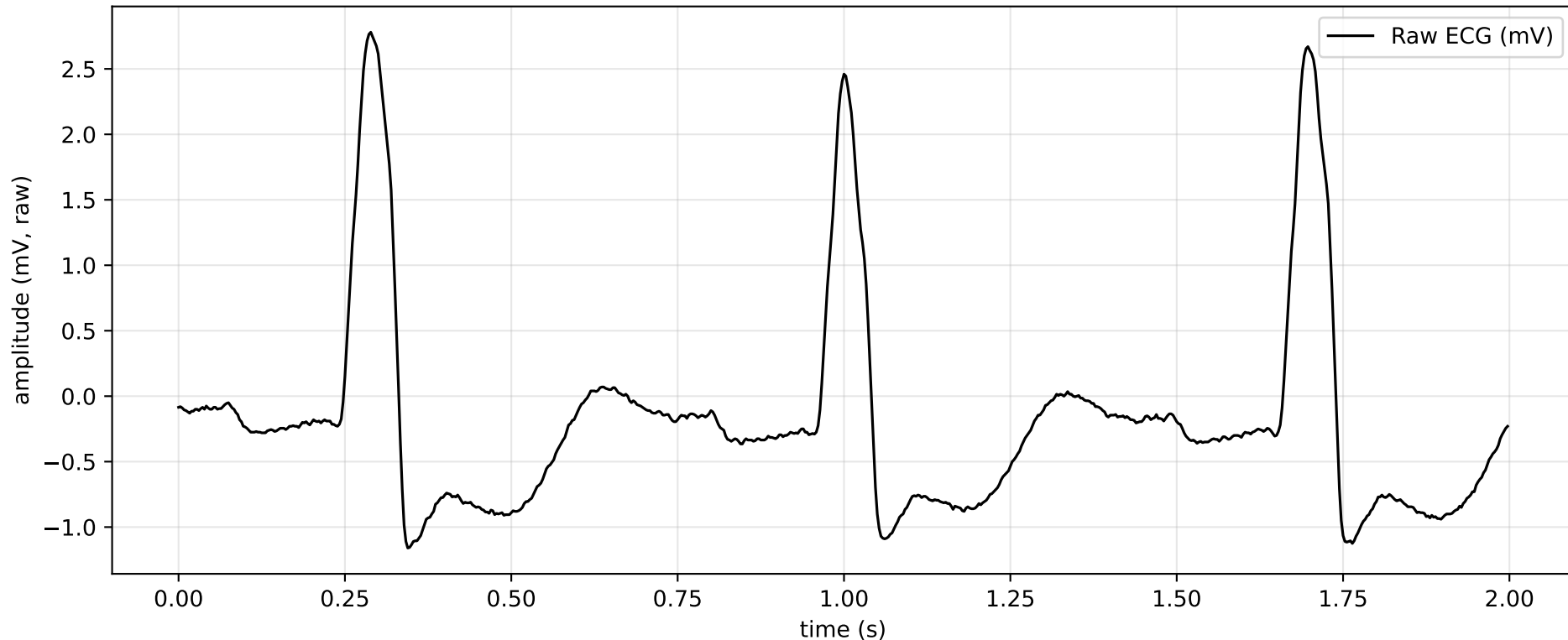
FN: Left bundle branch block (L)



Symbol	L
Type	Left bundle branch block
Source	record 207, sample 113694
Residual	0.0433
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	FN ✓ correct

NORMAL

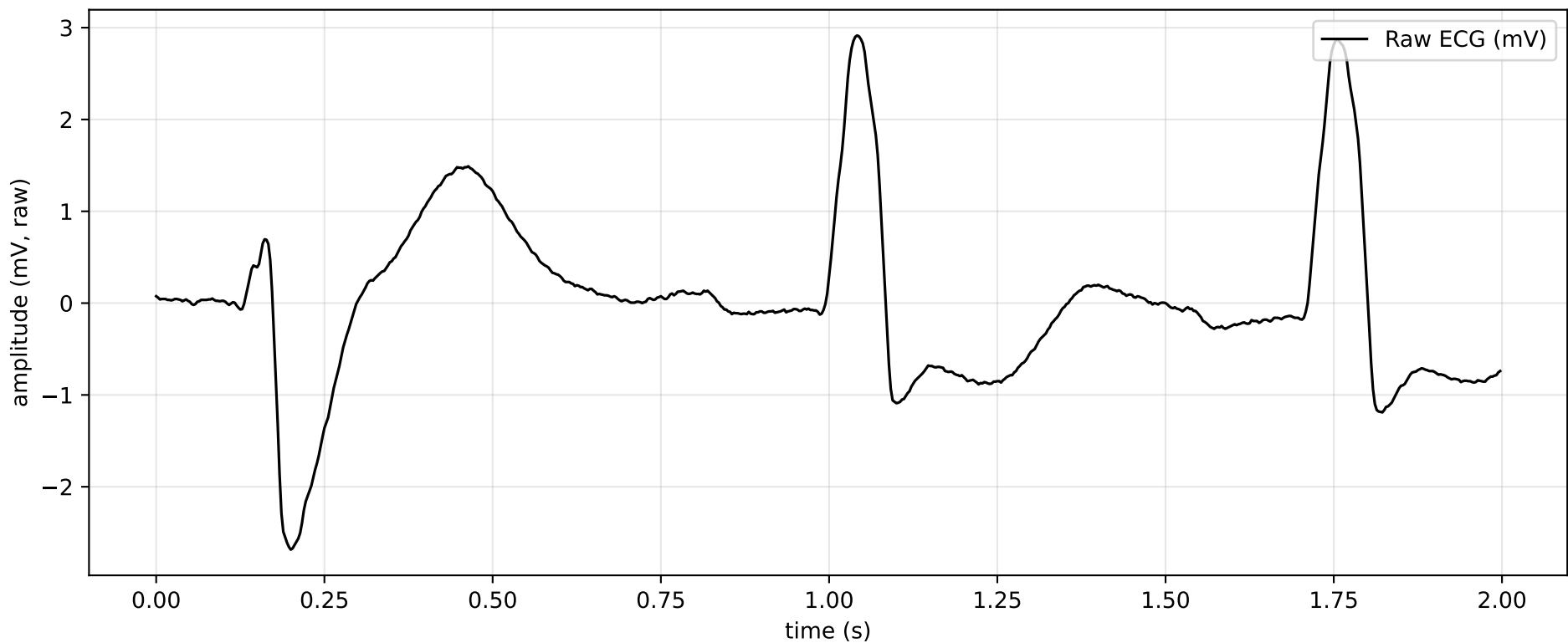
FN: Left bundle branch block (L)



Symbol	L
Type	Left bundle branch block
Source	record 109, sample 619904
Residual	0.0433
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	FN ✓ correct

NORMAL

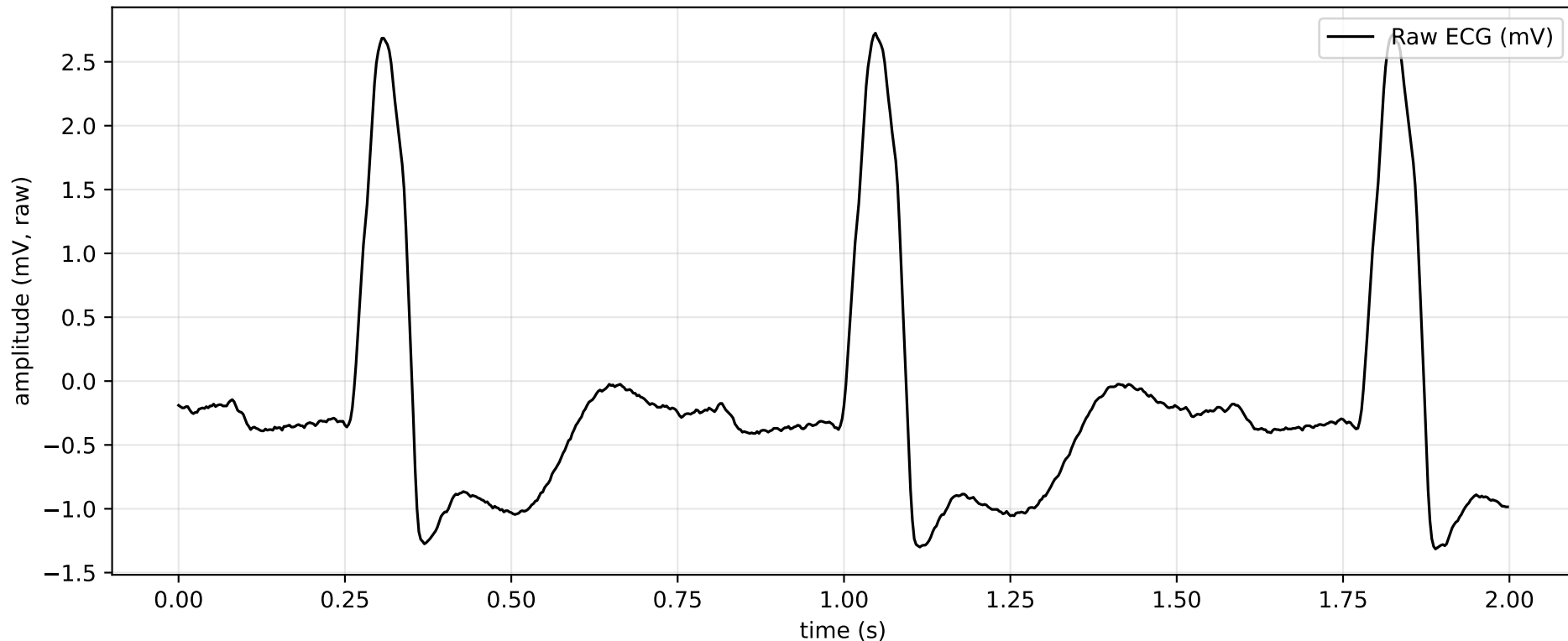
TP: Left bundle branch block (L)



Symbol	L
Type	Left bundle branch block
Source	record 109, sample 583956
Residual	1.7865
Threshold	0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	TP x mismatch

ANOMALY

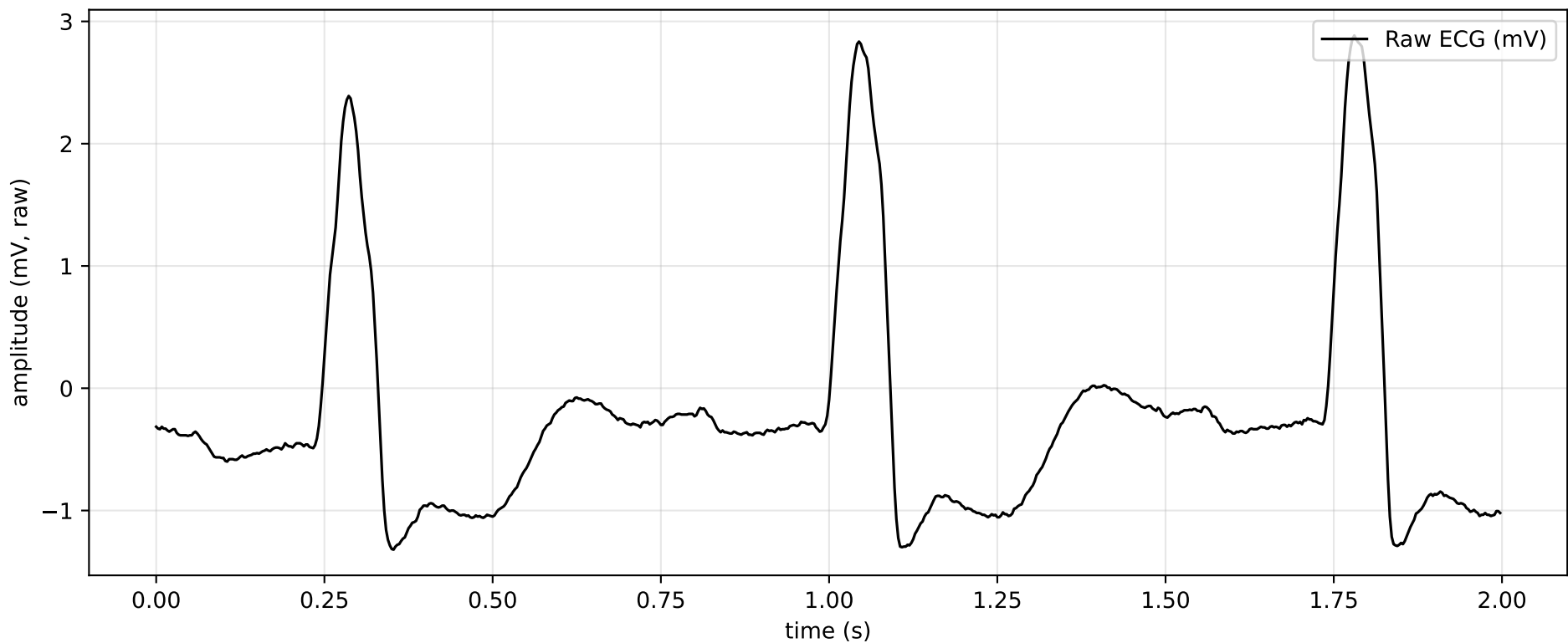
TP: Left bundle branch block (L)



Symbol	L
Type	Left bundle branch block
Source	record 109, sample 625192
Residual	1.6827
Threshold	0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	TP x mismatch

ANOMALY

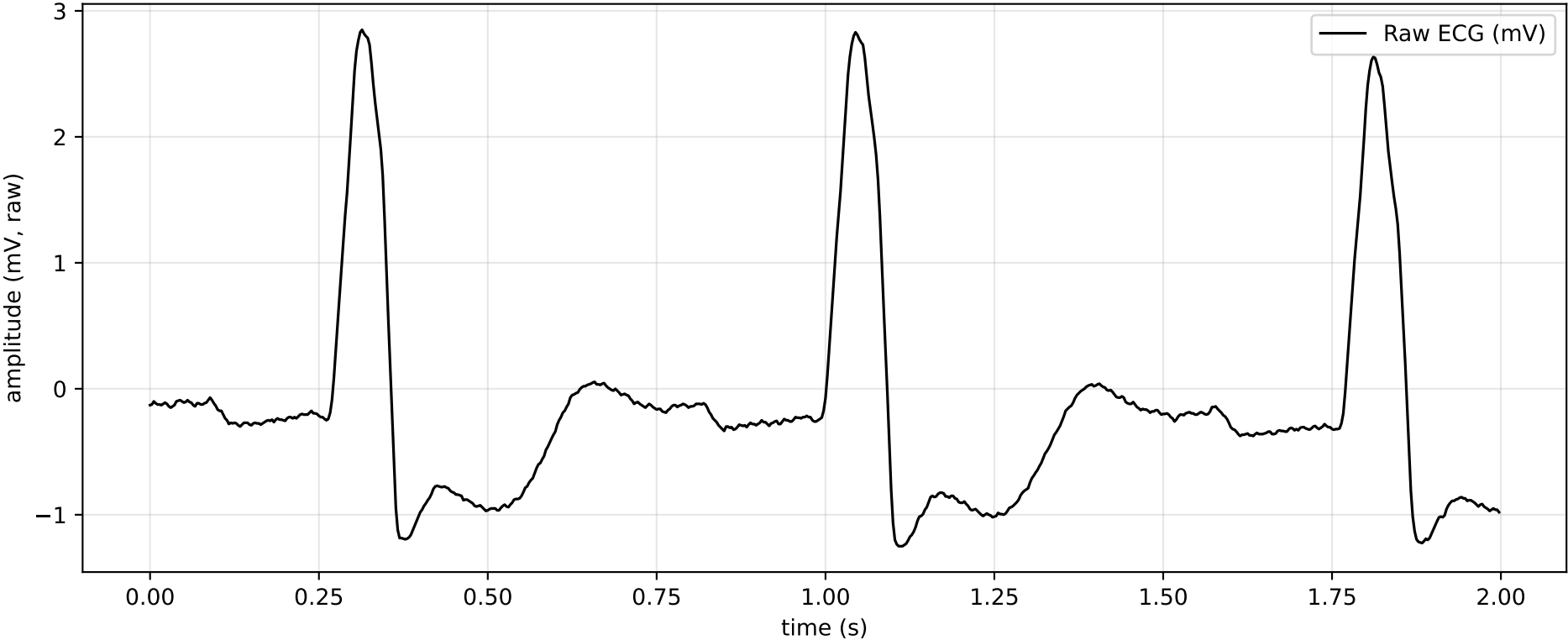
TP: Left bundle branch block (L)



Symbol	L
Type	Left bundle branch block
Source	record 109, sample 604399
Residual	1.6012
Threshold	0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	TP x mismatch

ANOMALY

TP: Left bundle branch block (L)

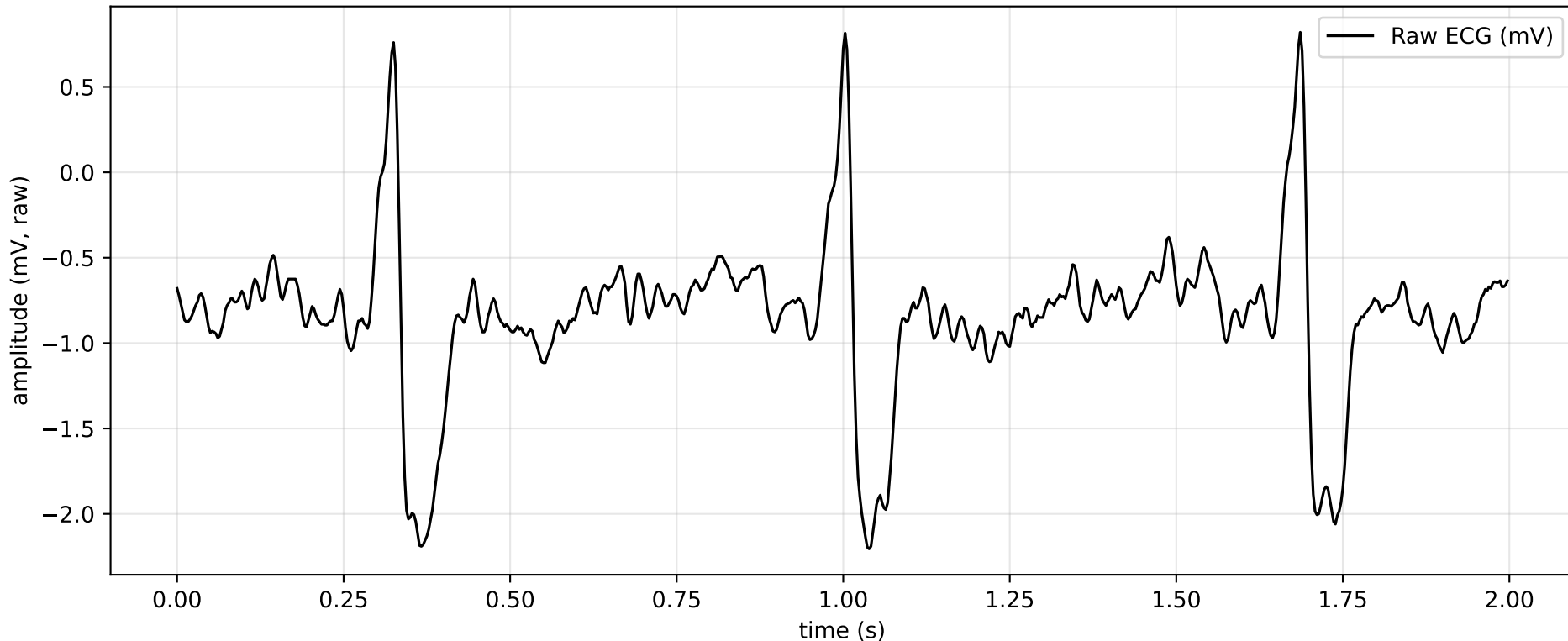


Symbol	L
Type	Left bundle branch block
Source	record 109, sample 601472
Residual	1.5828
Threshold	0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	TP x mismatch

ANOMALY

Section: R - Right bundle branch block

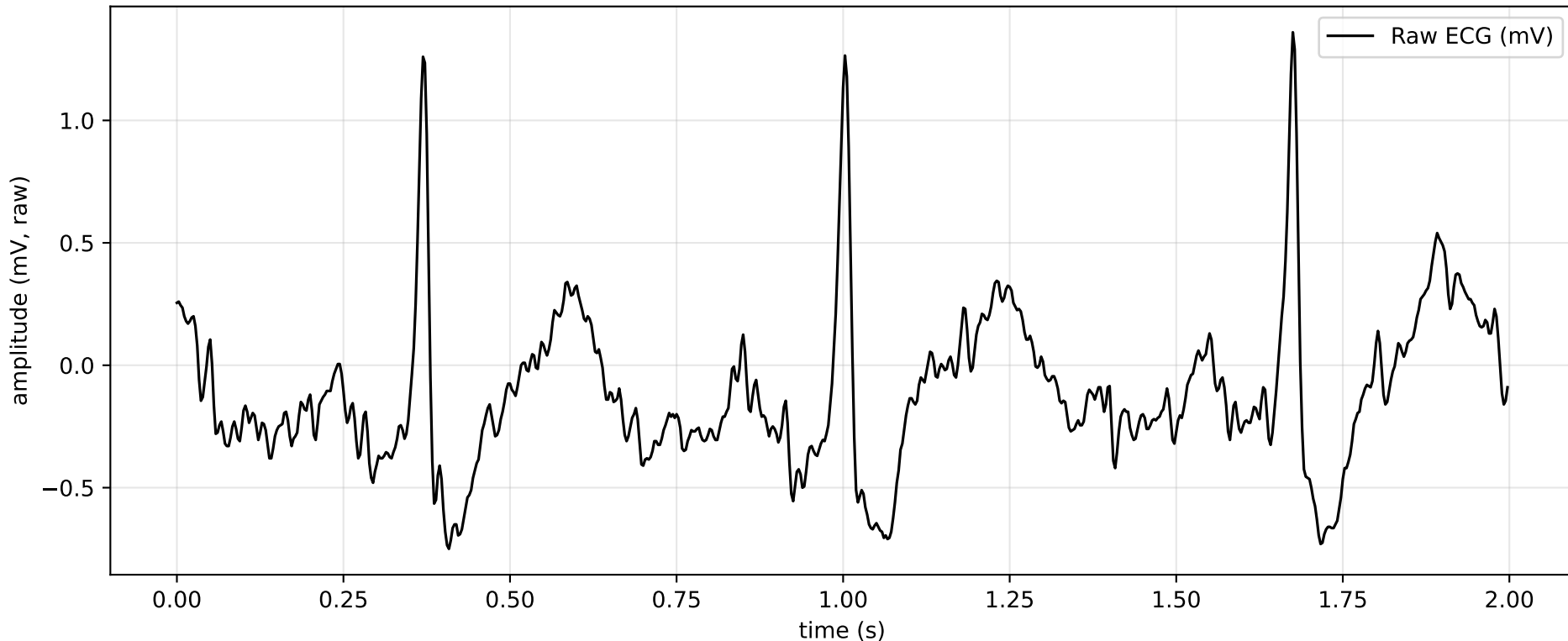
FN: Right bundle branch block (R)



Symbol	R
Type	Right bundle branch block
Source	record 118, sample 592155
Residual	0.0434
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	FN ✓ correct

NORMAL

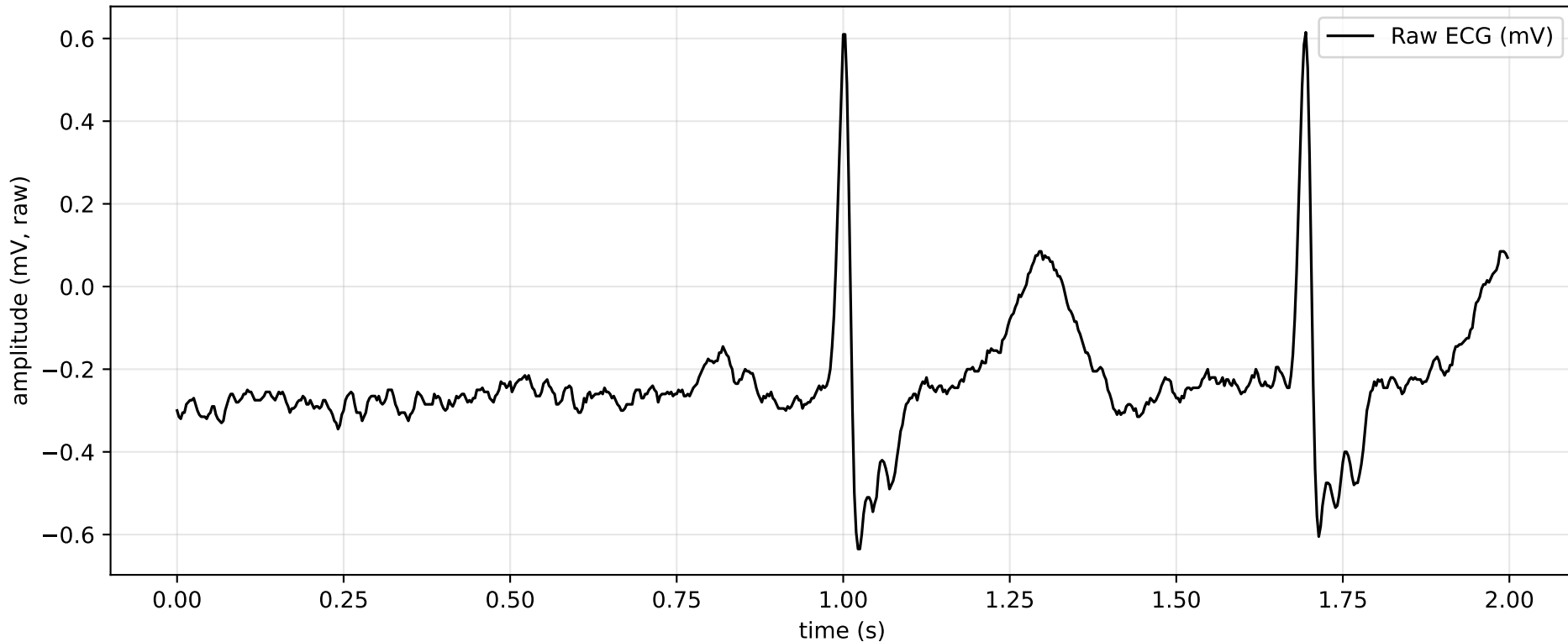
FN: Right bundle branch block (R)



Symbol	R
Type	Right bundle branch block
Source	record 212, sample 221336
Residual	0.0433
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	FN ✓ correct

NORMAL

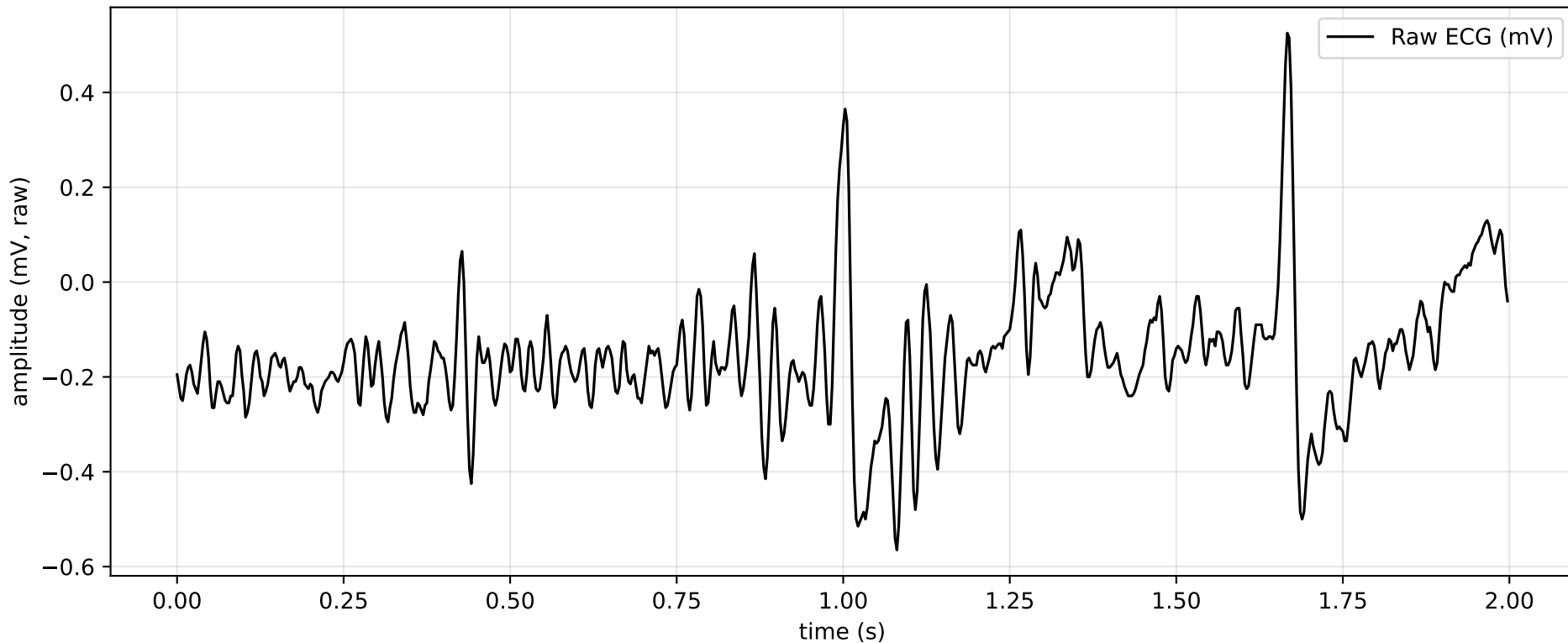
FN: Right bundle branch block (R)



Symbol	R
Type	Right bundle branch block
Source	record 232, sample 630475
Residual	0.0433
Threshold	0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	FN ✓ correct

NORMAL

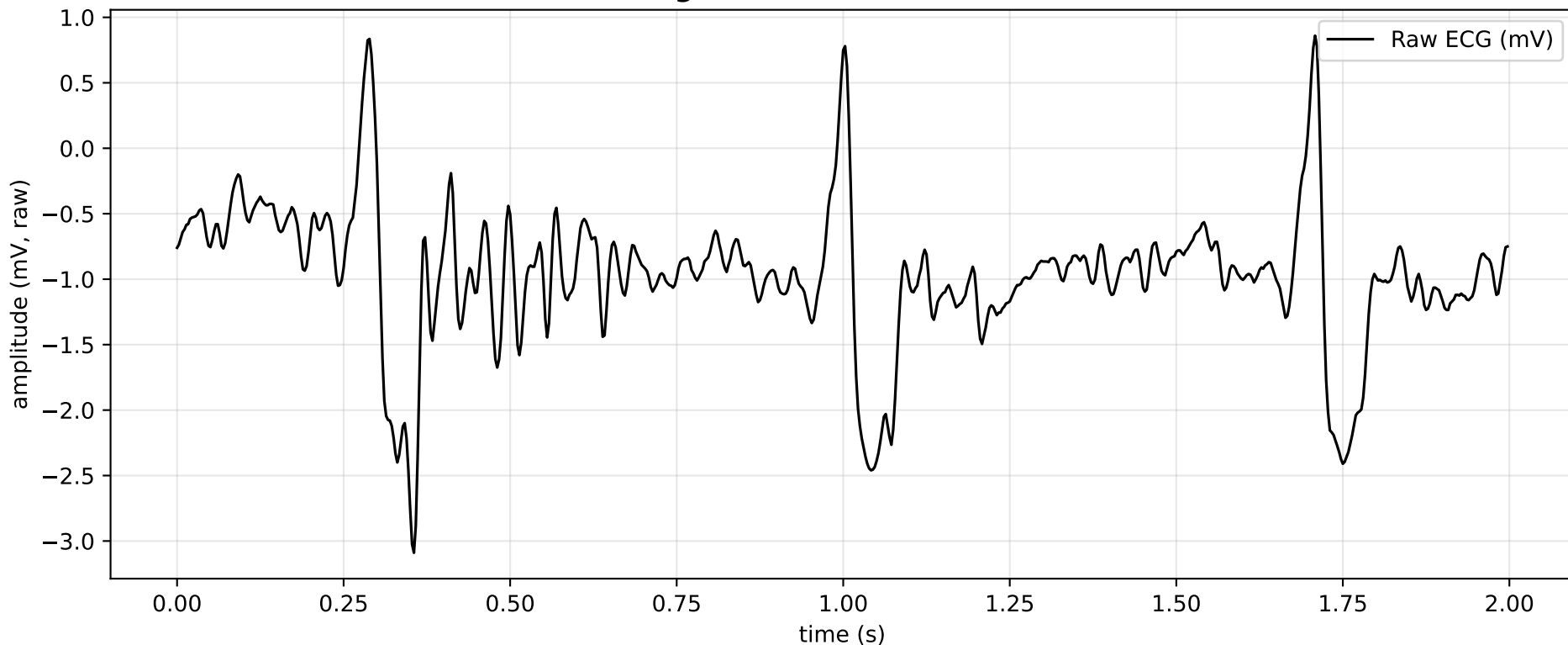
TP: Right bundle branch block (R)



Symbol	R
Type	Right bundle branch block
Source	record 232, sample 22726
Residual	0.2474
Threshold	0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	TP x mismatch

ANOMALY

TP: Right bundle branch block (R)

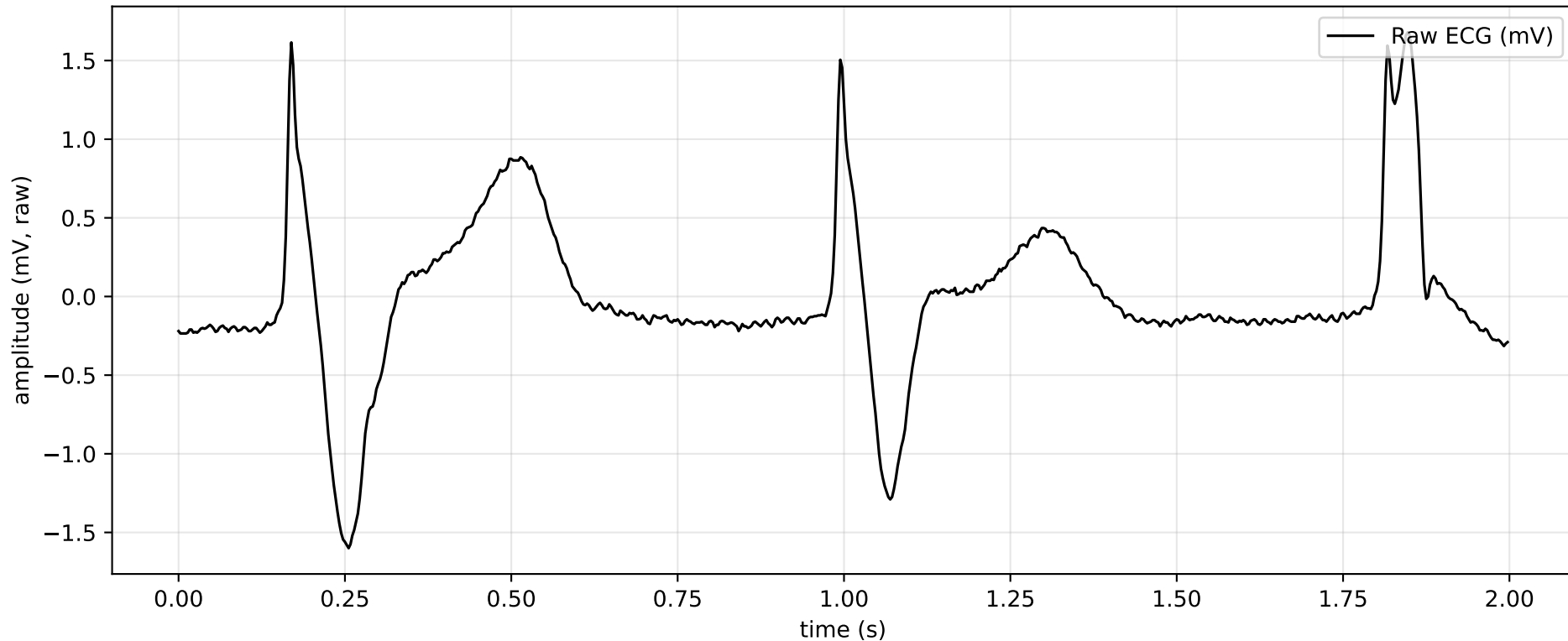


Symbol	R
Type	Right bundle branch block
Source	record 118, sample 621965
Residual	0.2204
Threshold	0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	TP x mismatch

ANOMALY

Section: / - Paced beat

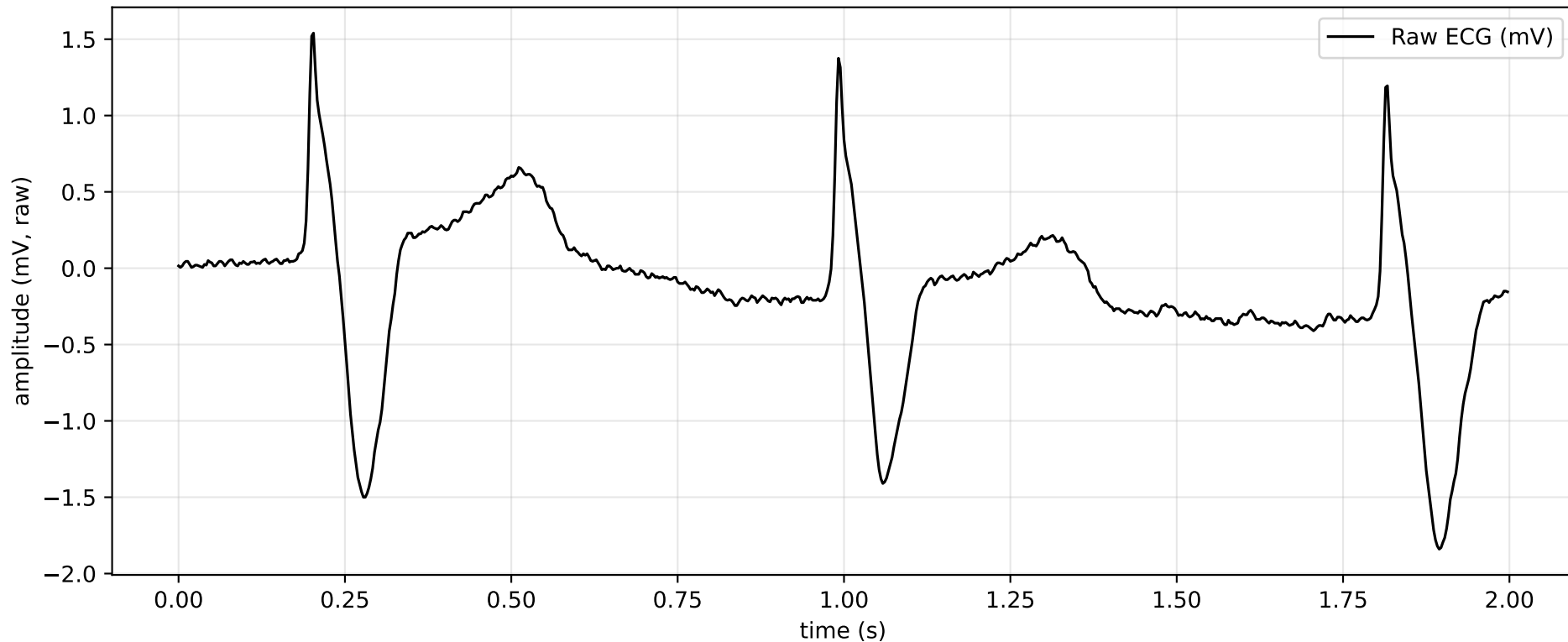
FN: Paced beat (/)



Symbol	/
Type	Paced beat
Source	record 217, sample 28465
Residual	0.0429
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

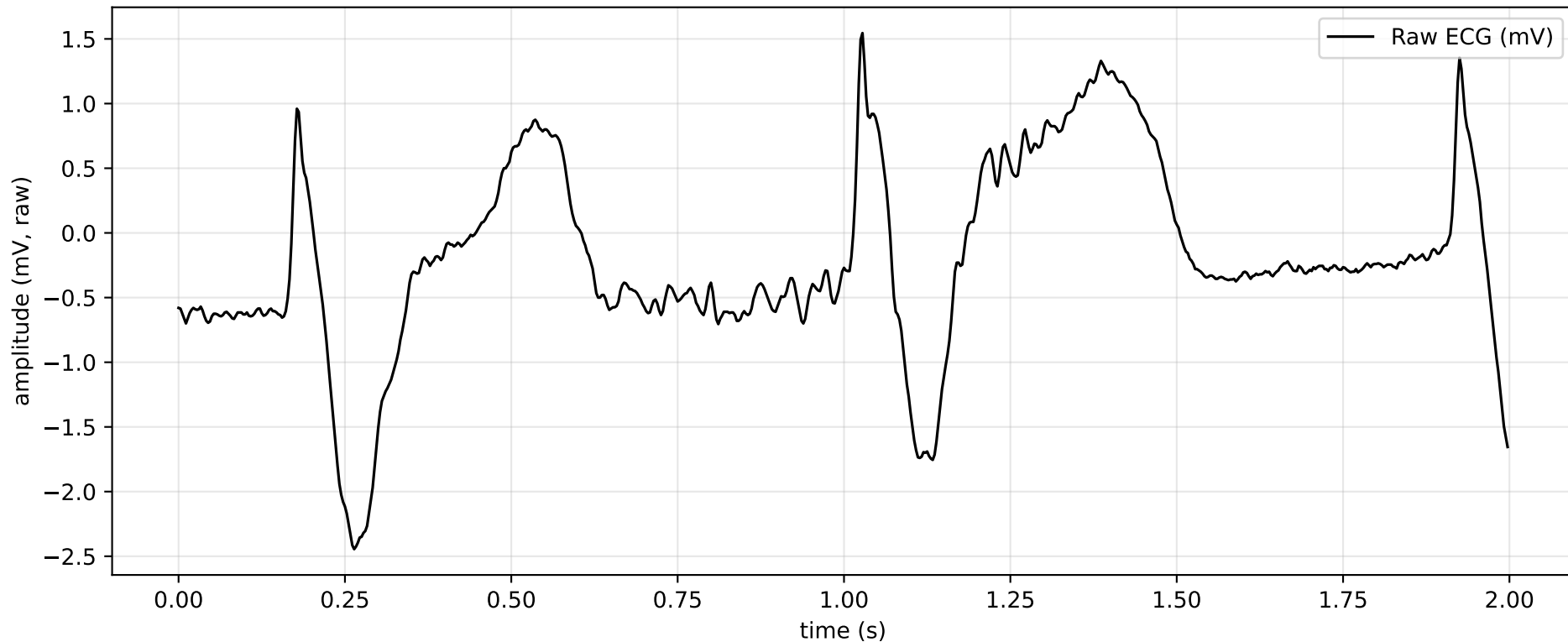
FN: Paced beat (/)



Symbol	/
Type	Paced beat
Source	record 217, sample 602890
Residual	0.0371
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

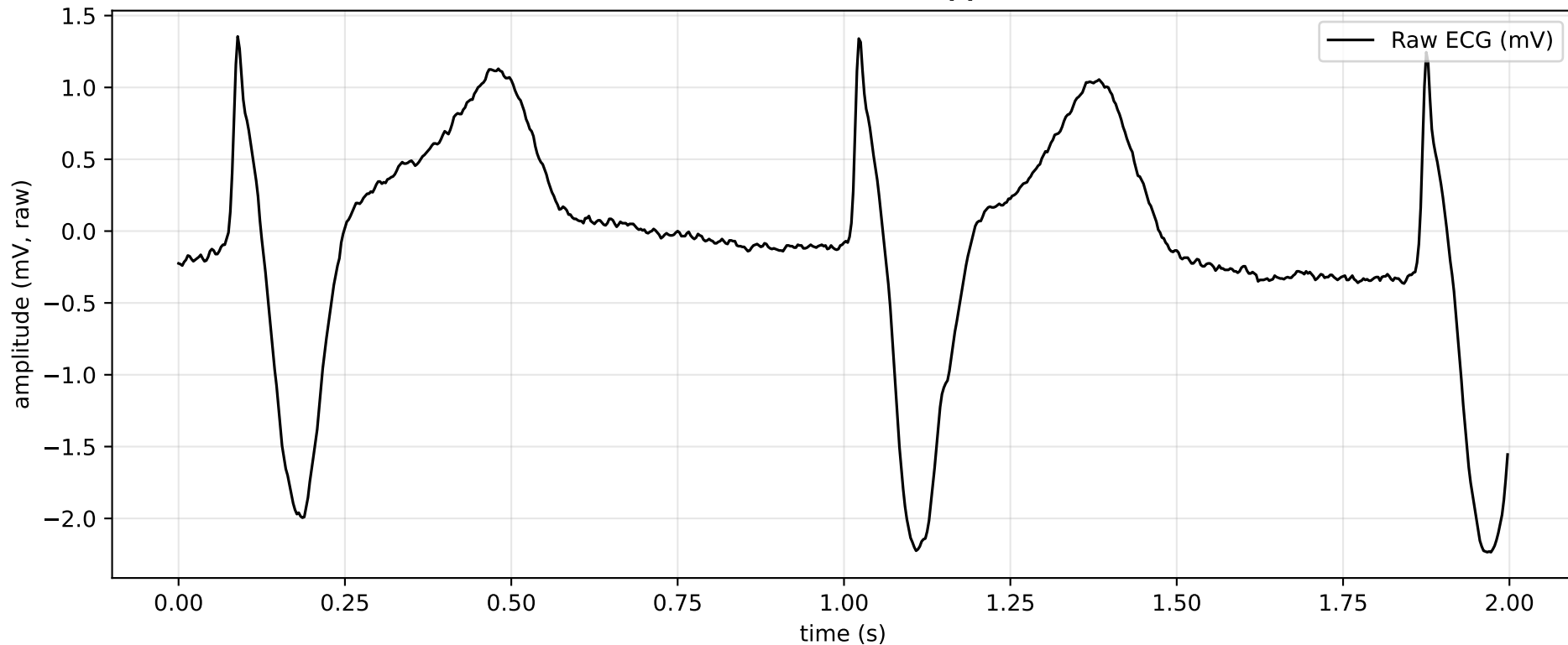
TP: Paced beat (/)



Symbol	/
Type	Paced beat
Source	record 217, sample 482286
Residual	0.2100
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

TP: Paced beat (/)

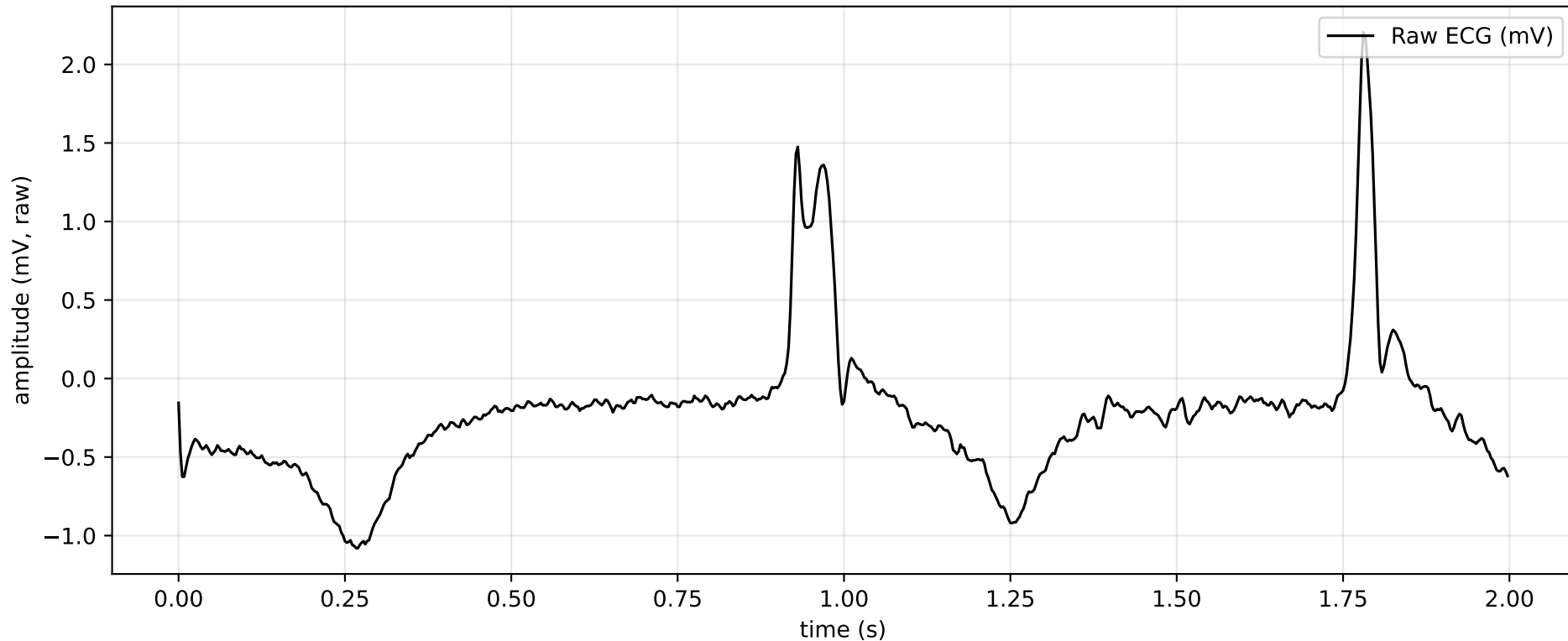


Symbol	/
Type	Paced beat
Source	record 217, sample 482947
Residual	0.1949
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY

Section: f - Fusion of paced and normal

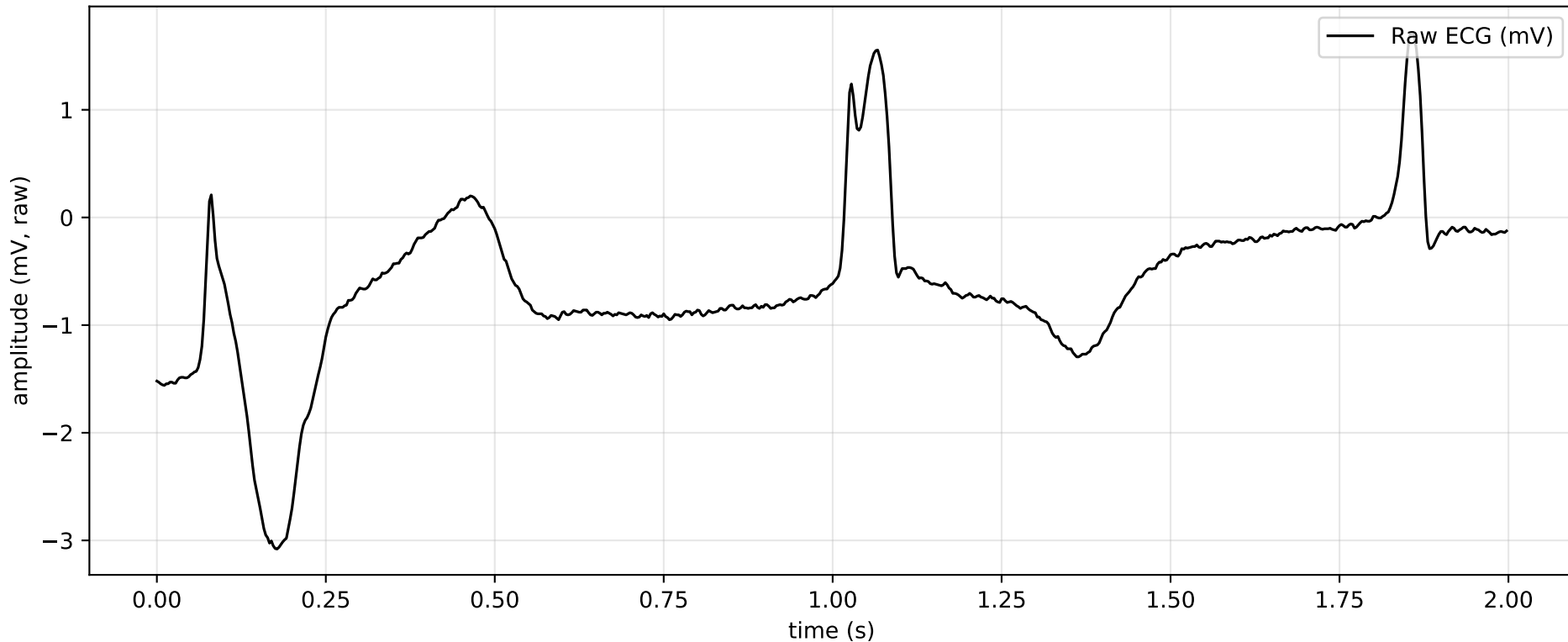
FN: Fusion of paced and normal (f)



Symbol	f
Type	Fusion of paced and normal
Source	record 217, sample 355577
Residual	0.0427
Threshold	0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN x mismatch

NORMAL

TP: Fusion of paced and normal (f)



Symbol	f
Type	Fusion of paced and normal
Source	record 217, sample 316353
Residual	0.1961
Threshold	0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP ✓ correct

ANOMALY